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Measuring Research Capacity Development In Healthcare Workers: a systematic review.

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Measuring Research Capacity Development In Healthcare Workers: a systematic review.

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ABSTRACT

Background: A key barrier in supporting health research capacity development (HRCD) is the lack of empirical measurement of competencies to assess skills and identify gaps in research activities. An effective tool to measure HRCD would help drive capable teams to undertake more locally-led research.

Aim: This systematic review identified tools measuring individual capacities of conducting research in Healthcare Workers (HCW).

Methods: 11 databases were searched (limited to English language) from inception to 16 January 2020. The first 10 pages of Google Scholar results were also considered. Two authors independently screened and reviewed studies using Covidence and performed quality assessments using the extraction log validated against the CASP checklist. The content method was used to define a meta-narrative analysis.

Results: The title and abstract of 7,474 unique records were screened and full texts of 178 were reviewed. A total of 16 papers were selected; seven quantitative studies, one qualitative study and five mixed-method studies. Tools with different levels of accuracy in measuring HRCD in HCW at the individual level were described. The Research Capacity and Culture Tool and the "Research Spider" tool were the most commonly defined. Other tools designed for ad hoc interventions with good generalizability potential were identified. Three papers described health research core competency frameworks. All tools measured HRCD in HCW at an individual level with the majority adding a measurement at the team/organisational level, or data about perceived barriers and motivators for conducting health research.

Conclusions: Capacity building is commonly identified with pre-post intervention evaluations without using a specific tool. This shows the need for a clear distinction between measuring the outcomes of training activities in a team/organisation, and effective actions promoting HRCD. This review highlights the lack of globally applicable comprehensive tools to provide comparable, standardised and consistent measurements of research competencies.

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ARTICLE SUMMARY

Strengths

- The review focuses on Health research capacity development (HRCD) for healthcare workers (HCW) that has been recognised as a essential element to overcoming global health challenges, especially in LMICs.
- One of the critical barriers in supporting HRCD is the lack of empirical measurement of competenceis in relation to the performance of research activities. Sixteen studies using or describing tools to measure HRCD were identified and analysed in this systematic review.
- This review produces evidence of the need for a standardised, comparable and consistent tool to measure individual HRCD to support a strategy for acquiring long term research competencies.

Limitations

- There is a lack of clarity on a common definition and terminology to identify HRCD which complicates the identification of relevant studies.
- Despite the presence of standards available for reporting qualitative or quantitative research there is a lack of studies which reference a standard reporting procedure.

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INTRODUCTION

In 2004 the Global Forum for Health Research highlighted the challenge for low and middle-income countries (LMICs) to have the capacity to perform effective and locally-led health research which address the major health problems affecting their own populations (1-3). Twenty years later, LMICs still carry 90% of the global disease burden, but only 10% of global funding for health research is devoted to addressing these persistent health challenges (4). Health research capacity development (HRCD) for healthcare workers (HCW) has been recognised as a critical element to overcoming global health challenges, especially in LMICs (5). For too long HRCD in LMICs has been documented through training programmes which enable local teams to perform externally sponsored trials, creating a false appearance of growth and generating dependence on foreign support (6, 7).

The process of progressive empowerment is usually referred to as capacity development (CD) (8). This term has been used in multiple areas and applied in different sectors to develop new or existing competencies, skills, strategies at a macro or individual level (9). In the field of health, research capacity development should support HCW in generating local evidence-based results to inform policy and improve population health. The three health-related Millennium Development Goals, and more recently the targets "B" and "C" of the Sustainable Development Goals, all support the adoption of new strategies to strengthen the capacity of HCW in all countries in performing their job and engaging in research (10-12). One of the critical barriers in supporting HRCD is the lack of empirical measurement of competenceis in relation to the performance of research activities. Existing frameworks and tools have been developed for a particular purpose in a particular context (13) (14). Others have identified barriers HCW encounter in becoming engaged in research or have monitored and evaluated targeted training activities (15). The need for an effective tool to measure health research capacities at the individual level, independently from any specific intervention or context, was the basis of the systematic review described. For this purpose, literature published from inception to 16 January 2020 was chosen and analysed to identify tools to measure individual capacities in conducting research in HCW across the world.

METHODS

The study was designed independently. Nevertheless, the PRISMA checklist (16) for reporting systematic reviews and meta-analyses and CASP checklist (17) on critical appraisal for qualitative studies was used to design the systematic review and specifically to refine the extraction log according to recognised guidelines.

Inclusion and exclusion criteria

Inclusion and exclusion criteria were defined in advance and documented using an adapted version of a SPIDER table (Table 1). The aim of the criteria was to identify tools which measure individual capacities in conducting research in HCW. The first element considered was the Sample, defined as the population included in a specific study. The primary sample of interest were HCW involved in research activities with any health-related professional profile working in healthcare. HCW delivering health services when research was not considered as the focus of the study were excluded. People involved in occupational health research were also excluded. Volunteers defined as people offering their services to support health activities with no specific training as health professionals were also excluded. Initially, only HCW working in LMICs were included, but this limitation was removed to identify any tool measuring HRCD in any setting. The second element of the table was the Phenomenon of Interest. The element was defined as: assessing HRCD; identifying tools, frameworks and templates designed to assess HRCD. A more comprehensive range of terms with identical or similar meaning to "assess" or "tool" or "development" was taken into consideration. Test searches resulted in excluding studies mentioning components that could be considered to assess, measure and

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'give evidence to' research capacity development, but were not presented in any capacity development context. In addition, since the concept of capacity development is widely applied to different settings, studies on areas unrelated to health such as "air pollution", "financial capacity" or "tobacco" were also excluded. The third element was the Design of the study defined as the nature of the intervention described. Qualitative, quantitative and mixed methods papers were included. Further criteria of eligibility included in the SPIDER table refers to the quality of the study (Evaluation) and the Research type.

Table 1 - ***SPIDER diagram - Inclusion and exclusion criteria***

SPIDER	*Inclusion*	Exclusion
Sample (population involved)	Healthcare workers (HCW) involved in research defined as: Health professionals involved in research activities at every step of research and with any health-related professional profile working in health centres.	Heath workers delivering health services when research is not considered People involved in occupational health research Volunteers defined as people offering their services to support health activities with no specific training as health professionals.
Phenomenon of Interest	Broadly, assessing health research capacity development. Specifically, analysing tools, frameworks, templates to assess and measure health research capacity development/building/strengthening. Key words: Tool – framework - assess – measure – give evidence - capacity development – capacity building – capacity strengthening - competency improvement – professional development - mentoring	Include specific components that could be considered as aiming to assess, measure and give evidence to research capacity development, but are not presented in any capacity development context. Non-health-related capacity development research such as "air pollution", "financial capacity", "tobacco" or other areas unrelated to health research emerging from the search refining.
Design (nature of the intervention described)	Qualitative literature including interview, focus groups, surveys, grounded theory, ethnography, interpretative phenomenological analysis, content analysis, thematic analysis. Quantitative literature including score systems, competencies measurement. Mixed methods papers combining qualitative and quantitative approach	Studies that do not meet the inclusion criteria for design.
Evaluation	Acceptable: meeting high and moderate score in quantitative assessment tool EPHPP and measure qualitative validity with COREQ evaluation method if applicable.	Studies that do not meet the inclusion criteria for evaluation.
Research Type	Qualitative, mix and quantitative methods	Do not have English language abstract

Information sources and search strategy

Eleven databases were searched from inception to 16 January 2020: Ovid Medline; Ovid Embase; Ovid PsycINFO; Ovid Global Health; Ebsco CINAHL; ProQuest Applied Social Sciences Index & Abstracts (ASSIA); ProQuest Sociological Abstracts; ProQuest Dissertations & Theses Global; Scopus; Web of Science Core Collection; and the WHO Global Index Medicus Regional Libraries. The first ten pages of results from Google Scholar were also screened. The search strategies used free-text terms and

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combinations of the relevant thesaurus terms, limited to English language publications only, to combine terms for capacity building, measuring, and health research. The "NOT" command was used to exclude papers about students, postgraduates, tobacco, air pollution, and a variety of other concepts to minimise the number of irrelevant results (Box 1 for a full set of search strategies).

BOX 1:

Database: Medline (Ovid MEDLINE® Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE® Daily and Ovid MEDLINE®) 1946 to present

Search Strategy:

---

1 Capacity Building/ (1965)
2 (capacit* adj2 build*).ti,ab. (5789)
3 (capacit* adj2 develop*).ti,ab. (3591)
4 (capacit* adj2 strengthen*).ti,ab. (924)
5 (competenc* adj2 improv*).ti,ab. (1460)
6 ((professional* adj2 develop*) and (competenc* or capacit*)).ti,ab. (1747)
7 1 or 2 or 3 or 4 or 5 or 6 (13649)
8 Mentoring/ (820)
9 mentor*.ti,ab. (13369)
10 (assess* or measur* or evaluat* or analys* or tool* or equip*).ti,ab. (9653076)
11 "giv* evidence".ti,ab. (3814)
12 framework*.ti,ab. (231138)
13 8 or 9 or 10 or 11 or 12 (9763562)
14 Research/ (196782)
15 clinical.ti,ab. (3158817)
16 (health* and research*).ti,ab. (337604)
17 14 or 15 or 16 (3588891)
18 7 and 13 and 17 (3433)
19 18 (3433)
20 limit 19 to english language (3346)
21 (student* or graduate or graduates or postgraduate* or "post graduate*" or volunteer* or communit* or tobacco or "climate change" or "air pollution" or occupational or "financial capacity" or informatics or "IT system" or "information system" or transport or "cultural competenc*" or disabili* or trauma).ti,ab. (1828113)
22 20 not 21 (1673)

Google Scholar – Screen the first 10 pages of results

Sorted by relevance:

("capacit* build*"|"build* capacit*"|"capacit* develop*"|"develop* capacit*"|"capacit* strengthen*"|"strengthen* capacit*"|"professional* develop*"|"completenc* improv*"|"improv* competenc*")("health* research*"|clinical)

https://scholar.google.co.uk/scholar?q=(%22capacit*+build*%22%7C%22build*+capacit*%22%7C%22capacit*+develop*%22%7C%22develop*+capacit*%22%7C%22capacit*+strengthen*%22%7C%22strengthen*+capacit*%22%7C%22professional*+develop*%22%7C%22completenc*+improv*%22%7C%22improv*+competenc*%22)(%22health*+research*%22%7Cclinical)&hl=en&as_sdt=0,5

Study selection

Two researchers, DB and ER independently screened and reviewed studies using the Covidence online tool for systematic review (18). In case of disagreement, DB and ER discussed the abstracts in question. After consensus on inclusion was reached, the full-texts of all included studies were rechecked for inclusion by DB and confirmed by ER.

Study analysis procedure

Data from selected papers were extracted, and quality assessments performed using an extraction log created ad hoc and validated against the CASP qualitative checklist. Macro areas of interest in the log

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were: general information on the paper such as author and title, main focus and study design. Explicit mention on the source of funding, conflict of interests and ethics approval were also recorded. A separate ad hoc section of the extraction log focused on collecting the characteristics of the tool used or described in each selected paper (Figure 1). If a definition of Capacity Development was given, the definition was collected as well as the presence of declared selection biases. The content method was used in defining a meta-narrative described in the discussion section of this article.

RESULTS

Database search and results screening

In December 2018, the first round of the search was performed in 11 different databases and in Google Scholar using the search strategy described in box 1. A total of 13,264 suitable records were found. A total of 6,905 duplicates were removed, resulting in 6,359 unique records for inclusion screening by title and abstract (Table 2), which was performed throughout 2019. In January 2020 an additional search for papers published or included in publication databases in 2019 was performed using the same search strategy and resulted in 15,775 papers and after removal of duplications, a total of 1,118 papers were found. These papers were then added to the 6,359 papers identified from the first search. The total number of suitable papers included for title and abstract inclusion screening, after finding three other duplications in Covidence, was 7,474 unique, relevant studies. Even if chronologically the first tile and abstract screening was performed between the first and the second round of search, the methodology followed in identifying relevant papers was identical for both sets of results and is described below.

Table 2 - ***Search Results***

	Search dates: 20-21/12/18	Search date: 16/01/2020
Ovid Medline (searched on 20/12/18)	1673	1937
Ovid Embase (searched on 20/12/18)	2344	2721
Ovid PsycINFO (searched on 20/12/18)	597	619
Ovid Global Health (searched on 20/12/18)	566	676
EBSCO CINAHL (searched on 20/12/18)	1376	1663
ProQuest Sociological Abstracts (searched on 20/12/18)	305	371
ProQuest ASSIA(searched on 20/12/18)	1463	2036
ProQuest Dissertations & Theses Global (searched on 20/12/18)	279	312
Scopus (searched on 21/12/18)	2230	2575
Web of Science – Core Collection (searched on 21/12/18)	1900	2195
WHO Global Index Medicus (searched on 21/12/18)	531	670
Total	13264	15775
Total after deduplication	6359	7469
16/01/2020: Unique to databases since 20-21/12/18		1118
TOTAL TO SCREEN		7477

The 7,474 unique relevant studies identified were uploaded to the Covidence online tool. Two researchers, DB and ER independently screened the studies including or excluding studies according to the SPIDER table criteria (Table 1). The Covidence system presented the divergent opinions between the two researchers and disagreements were resolved with a dedicated discussion session. A total number of 7280 studies were considered irrelevant. The full text of the remaining 178 were reviewed. Reasons for exclusion were identified by streamlining the SPIDER table criteria into three main criteria: wrong setting, irrelevant study design and wrong focus of the study. A reason for exclusion was assigned to each paper. All 178 studies described some form of activity to measure the transfer of

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competencies related to performing health research. Thirty of them were excluded due to being a literature review on a different aspect of health research or because they described a general perspective on the topic of health capacity development without offering any specific measurement or without reference to research. In addition, 42 studies were excluded because of the wrong setting, since competencies were measured at the level of research institutions or within a specific network. Additional 90 studies were excluded because the study design did not match the inclusion criteria; of these 90 studies, 38 studies described the use of a measurement tool tailored around the context (e.g. specific profession, intervention or setting) and not at the individual level, with the remaining 34 studies excluded since there was no mention of any specific tool to measure HRCD. The final 18 papers reported the use of an evaluation tool, but the tool was an *ad hoc* pre-post intervention questionnaire with low potential of applicability in a context different from the one described in the paper. A total of 162 studies were therefore excluded, leaving 16 studies for this review (Figure 2).

Analysis of the findings across the selected papers

A total of 16 studies met the inclusion criteria set for this systematic review (19-34). The 16 articles were analysed using the extraction log created ad hoc and validated against the CASP qualitative checklist.

The results are summarised in Table 2. None of the papers were published before 2006 and 9 of them after 2014 (20, 21, 23-26, 31, 33, 34). The majority (n = 13) applied a tool in high-income settings (19, 20, 22-24, 26-32, 34). Seven papers described the use of tools in Australia (20, 22, 24, 26, 28, 29, 34), 3 in low and middle-income countries (one in Ghana, Kenya, Malawi, Sri Lanka (25), one in the Pacific Islands (21) and one in the Philippines (33)), one in Europe (Norway) (19), one in the US (32), and one measured HRCD in a group linked to a specific intervention located in multiple areas of the world. Three of them described the creation of a tool without applying it to any specific context (23, 30, 31), but they were all designed by research groups in high-income countries (one in the US and two in the UK).

All of the selected studies applied quantitative, qualitative and mixed-method analyses. The preferred approach (n=7) was to generate quantitative data using an HRCD tool (20, 24, 26-28, 33, 34). One-third of the studies (n = 5) used a mix method approach (19, 21, 22, 25, 29); quantitative tools were associated with semi-structured interviews or in some cases qualitative questions were added to the questionnaire. The three studies describing the creation of a tool were not analysed under this methodological category.

Of the 16 selected studies, 3 used the term “capacity development” (23, 25, 30), and 2 included a definition of the concept (25, 30). Seven papers used “capacity building” (20-22, 24, 26, 28, 31), of which 4 also included a definition (20, 22, 24, 28). In 2 papers, the definition of capacity building was associated with the definition of research culture (20, 22). Two additional papers used alternative generic terms like “research capacity” (33) or “research self-efficacy” (32). Four papers did not refer to any specific term and therefore no definition was given (19, 27, 29, 34).

Five of the 16 selected papers openly declared no conflict of interests (22-24, 28, 31). Eight stated the source of funding used to carry out the activities described (19, 21, 23, 27-31). The number of participants in the studies varied from 28 enrolled participants for a qualitative study (21) to 3500 users of an online measurement tool (27).

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Analysis of the tools from the selected papers

The tools described or used in the 16 selected papers varied in nature, length and applicability. In general, even when there were similarities, each paper described a different perspective on the use of a tool. Nevertheless, it is possible to group them into categories. Four papers applied a questionnaire-type tool to assess research competencies and skills (19, 21, 25, 33). The length of these questionnaires varied from 19 (21) to 59 (19) health research capacity related questions, with the addition of open-ended qualitative questions in two studies (19, 21), and a structured interview in another study (25).

Three studies (22, 24, 34) used, with a range of adaptations, the Research Capacity and Culture Tool (RCC) and one study (20) revised the RCC tool into a Research Capacity and Context Tool (RCCT) referencing RCC as a primary source. Another recurrence in the papers was the use of the "Research Spider" tool (28, 29). Again, adaptation to the context had been made to the original tool, and in one case (29), the tool was used as a base for qualitative research on HRCD. Two additional papers described tools designed *ad hoc* to measure the impact of an intervention (CareerTrac (27) and Cross-Sectional Electronic Survey (26)). These last two papers were not excluded under pre-post intervention since the action was wider, at a programme level, and the tool used to measure HRCD was the main focus of the paper. Furthermore, another paper described a tool for a specific category of HCW (Nursing Research Self-Efficacy Scale -NURSES) (32). Three papers (23, 30, 31) focussed on the creation of a new tool and described the process of identifying a set of competencies required to run health research. The outcome of two of them was defined as a "core competency framework" (23, 31). The third defined the outcome of the analysis as a "set of indicators" (30).

In terms of the target population, the identified tools aimed to measure HRCD in a range of different HCW professions. One-third of the papers (n=5) focused on measuring HRCD on Allied Health Professionals (AHPs) (20, 22, 24, 26, 34). Nurses were the main focus in 2 other studies (19, 32), and 4 studies applied a tool to a range of health professions (ranging from laboratory scientists to data managers) (21, 25, 28, 29). Two other papers focused on groups linked to a specific intervention (27, 33). All 16 papers included, alongside HCW, representatives of technical professions in health such as managers, directors, faculty members, consumer organisation representatives. In the case of the 3 papers describing the creation of a new tool, they suggest that these tools would be applicable to all research roles (23, 30, 31).

As per inclusion criteria, the main level of measurement of the tools was at the individual level. Seven papers only measured HRCD at the individual level (19, 23, 28, 29, 31-33). Three papers added to the individual level of measurement by including information on the perceived barriers in performing health research (21, 26, 29); of these three, two also focused on understanding what motivates HCW to become involved in health research (26, 29). The five studies, which used the RCC tool and its variants, included the measurement of HRCD not only at the individual level, but also at the team and organisational level (20, 22, 24, 25, 34). One paper described the creation of a tool designed to be used at the organisational level, but embedded a measurement of HRCD at the individual level as well (30).

The most common way a selected tool was validated was by referencing the main paper that described the selected tool and its validation process (n = 6) (23, 28, 29, 32-34). This was the case for some of the *ad hoc* questionnaires (23, 33, 34), of the "Research Spider" tool (28, 29), and of the NURSES tool (32). Papers which described an original process or used modified versions of an original tool validated the tool through a contextual validation process described in the paper (21, 22, 24, 25, 31). These validation processes included a consultation of a panel of experts (22, 24, 31) or a reiterative process

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of validity (21, 25). One paper stated that the tool used was a validated tool without referencing the process/tool (20).

Overall, only two papers (23, 31) focussed specifically on tools to measure HRCD on a wider level, without linking the measurement to a specific group or a geographical area which was done in the majority of papers (19, 24, 25, 28, 29, 33). In four cases the tools described were adapted to identify determinants or barriers of HRCD in a defined setting (20, 30, 34) or to promote HRCD in relation to a specific disease or research topic (21). In other cases, the papers focussed on a tool aiming to assess the impact of specific interventions or programmes on HRCD (26, 27).

DISCUSSION

Summary of evidence

This systematic review aimed to identify tools which measure individual capacities in conducting research in HCW; the 16 included articles (19-34) demonstrated that tools to measure HRCD in HCW despite being limited in number, are available. In most cases the identified tools do not originate from the need to measure and foster HRCD as a necessary strategy to promote research capacity. There is, therefore, a need to design more comprehensive tools which are globally applicable and able to provide comparable, standardised and consistent measurements of research competencies.

The importance of measuring HRCD has only been recognised recently (15). From the date of publication of the identified papers show, the contribution that health research can offer in capacity development at a personal level only begun in the first decade of this new millennium. Almost half of the selected papers (n= 7) refer to studies whose data have been collected after 2014 (20, 21, 24, 26, 31, 33, 34). Of note is the high number of new publications which were retrieved from the academic databases (1,118 papers) when the search strategy was re run in 2019.

The questionnaire was the most commonly used method for assessing research skills and competencies. Almost two-thirds of the papers (n = 10) (19, 20, 22, 24, 26, 28, 29, 32-34) based the measuring system of different research skills at a personal level using a 5 point Likert scale (n = 6) (19, 26, 28, 29, 32, 33) or a 10 point scale (n = 4) (20, 22, 24, 34). This choice highlights the need for a validated quantitative tool based on a set of competency related questions that can bring standardisation, comparability and consistency across different roles and contexts. However, the extensive use of mixed methods, combining quantitative questionnaires with other qualitative instruments, reflects that HRCD depends on a complex series of components that need to be identified both qualitatively and quantitatively.

By not limiting the selection of articles for this review to those tools used in low and middle-income countries (LMICs), revealed that most of the tools identified were used in high-income settings. It is important to note that excluding pre/post-intervention assessments significantly reduced the inclusion of studies performed in LMICs. This finding highlights that although health systems in LMICs may benefit from providing evidence for HRCD (5), they are rarely the focus of the HRCD literature. Most of the measurements of HRCD in lower-income settings appears, in fact, to be narrowly linked to the measurement of the effectiveness of training offered for a specific study or limited to a particular disease. Even when the perspective is broader than a particular study, it is mostly limited to the evaluation and sustainability of training programmes and not linked to a plan of career progression and research competency acquisition. More attention should therefore be given in creating tools which are able to measure, support and promote long-lasting research capabilities in the perspective of professional growth for HCW.

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Three essential findings of this systematic review support a change in the perception of HRCD and the tools needed to measure it. Firstly, many of the excluded papers (42 out of 162 excluded papers from the last round of analysis) focused exclusively at the institutional level of measuring research capacity. This is mostly because training interventions are designed to prepare a team to run a study and rarely to promote individual HRCD (1, 35, 36). In some cases, the measurement via a tool is also an exercise to demonstrate the investment in training activities for reporting purposes (37, 38). It is therefore important to start promoting a more effective research culture which is independent from specific diseases or roles. This progression could be achieved by championing systems which measure the changes in research capacities at a team and personal level using a globally applicable tool. Most of the tools excluded were evaluation tools designed for, or used in a specific setting and thus not suitable for a comparable, standardised and consistent analyses of long term research competency acquisition strategies.

Secondly, papers that focused on measuring HRCD at the individual level confirmed that research is seen as an opportunity to learn the cross-cutting skills needed in healthcare. A defined set of standardised competencies required to appropriately run research could be used to measure an individual, team and organsaition's ability to conduct research. This was the focus of two papers (23, 31) which identified a framework of core competencies. Most of the tools (n = 7) were designed to be applied to a wider variety of health professionions (21, 23, 25, 28-31). HRCD can be accessed at different entry points depending on the specific job title, but the set of skills acquired is common and shared among the research team (1). The approach on assessing these inter-related competencies should therefore be global and not role or disease-based (39). The measurement at an individual level is essential to promote a consistent and coherent career progression for each person and role (40). However, the overall capability in running research programmes should be measured at a team level where all roles and competencies complement each other, skills are made visible, and measurable as a whole against an overall competency framework . Individual and institutional/team levels are therefore, two aspects of HRCD that grow together supported by a common comparable, standardised and consistent tool.

Thirdly, the lack of a standard definition for HRCD can lead to post-training evaluations being categorised as HRCD activities. Although pre/post training evaluations are important, it might be helpful to define what a 'structured action' is to promote HRCD. As previously mentioned, the term "capacity development" is not universally used, with many synonyms such as "capacity building" or "capacity strengthening", creating the possibility of different interpretations. Furthermore, inconsistent terminology was found in describing activities in support of HRCD that in reality, were very similar (e.g. workshop, training, course). Steiner *et al.* (41) suggest that there should be a standard definition in the context of educational capacity development. This suggestion, alongside a common taxonomy to describe health professions, would support the identification of HRCD as a defined process with specific characteristics and not with a general effort for research training.

The most common tool identified in this review was the Research Capacity and Culture Tool (RCC) (20, 22, 24, 34). The RCC consists of 52 questions that examine participants' self-reported success or skill in a range of areas related to research capacity or culture across three domains including the organization (18 questions), team (19 questions), and individual (15 questions). The RCC tool includes questions on perceived barriers and motivators for undertaking research. The respondents of the RCC are asked to rate a series of statements relevant to these three domains on a scale of 1–10, with one being lowest skill or success level and ten being the highest possible skill or success level. It represents

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a good example of a comprehensive tool. As confirmed by the review findings, a potential limitation is its application mainly in an Australian context and almost exclusively to measure HRCD in AHPs (22, 24, 34). The generalizability of the tool should thus be confirmed. Nevertheless, the RCC represents a strong example of how having a tool refined around a context, and a specific health profession can be an incentive in measuring HRCD.

Another tool highlighted by this review was the "Research Spider" tool (28, 29, 42). This tool collects information on individual research experience and interest in research skill development in ten core areas. These include "writing a research protocol", "using quantitative research methods", "publishing research", "finding relevant literature", and "applying for research funding". In each area, the level of experience is measured on a five-point Likert scale, from 1 (no experience) to 5 (high experience). The primary aim of the "Research Spider" is to be a flexible tool. This flexibility is confirmed in two studies (28, 29) which used the "Research Spider", with one (28) using it as the main measurement, and the other (29) as a quantitative base for qualitative semi-structured interviews. The advantage of the "Research Spider" tool is that it provides a visual overview of personal research competencies. However, the limited number of measurement areas (n = 10) makes the tool a good initial evaluation instrument, but does not offer a specification of the sub-skills of each area.

A critical mention should be reserved for the two papers which described the creation of a comprehensive research core competency framework (23, 31). Despite no specific tool being described and the competency scores being visualised by using a spider diagram, these studies present the most accurate overview of the skills required in running research programmes related to health. As mentioned before, a tool which applies a scoring system to the list of competencies identified by these frameworks has the potential of being widely applicable and reliable. This wide applicability and the absence of explicit biases in measuring research skills improvement, can foster a more robust approach to research in health. The measurement of HRCD unrelated to specific interventions would maximise the benefit of research at every level. At a personal level, it would clarify a potential career progression path highlighting possible gaps; at the team level, it would support a multidisciplinary approach to health challenges; and at an institutional level, the measurement of HRCD would make the know-how generated by the international scientific community accessible to a broader group of local health workers. Overall, health practice at a global scale would benefit from the incentive of getting involved in research derived from measuring the impact of it on improving competencies. Thus positive outcomes of measuring HRCD could place the issue of universal transferability, and applicability of research methodology and results at a higher level of priority in the design of health research projects.

Limitations of the Systematic Review

Methodological limitations should be recognised for the present systematic review; firstly, there is a lack of clarity on a common definition and terminology to identify HRCD which complicates the search strategy. A long reiteration process was necessary during the database search strategy to try and include all the possible variants used to define "tool", "capacities" and "development". Despite this effort, some studies may have been missed. Secondly, there was a lack of studies which referenced a standard reporting procedure, despite the presence of standards available for reporting qualitative or quantitative research (43-45) as well as for mixed-method research (46). Other limitations typical for reviews may also apply. Despite the attempt of being as comprehensive as possible, some sources might not have been detected due to the challenge in finding all the relevant grey literature, and the restriction to English sources.

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CONCLUSIONS

Sixteen studies using or describing tools to measure HRCD were identified and analysed in this systematic review (19-34). Identifying capacity development with pre-post intervention evaluations or to generically evaluate capacity development without using a tool was common. Furthermore, the majority of papers were not included due to measuring HRCD at an institutional level and not at an individual level. This finding highlights the need for a clear distinction between simply measuring training activity outcomes in HCW and effective action promoting HRCD for HCW. Further research on harmonisation and clarification of terms related to HRCD would contribute to solving this issue.

Identified tools were analysed on the basis of their frequency within the selected papers. The most recurrent tool described was the Research Capacity and Culture Tool (RCC) based on a 1 to 10 scoring system in 3 areas of research linked to institutional, team and individual level (20, 22, 24, 34). The second most recurrent tool was the "Research Spider" tool which assesses ten core areas of research scored on a 1 to 5 Likert scale (28, 29). A variety of other tools, mostly questionnaire-based, were identified, and in most cases, a broader applicability than described in the specific context of the paper may be possible. Two frameworks systematising research core competencies were identified (23, 31). The potential of tools derived from these frameworks could be significant. The competency areas in conducting research identified by the two frameworks mostly coincide, but the depth of the evaluation in each area varied in each tool. The applicability of each tool depends, therefore, on the context and on the level of accuracy needed

Future directions for HRCD include the design of a standardised, comparable and consistent tool to measure individual HRCD not linked to training evaluation, but support a long term research competencies acquisition strategy. In addition, the harmonisation of definitions and terminologies used in identifying HRCD actions and processes could facilitate standardisation and comparability of HRCD strategies.

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Data sharing:
All data relevant to the study are included in the article. Data are available upon reasonable request.

Competing interests
The authors declare that they have no competing interests.

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Review registration
The protocol for this systematic review has been registered with PROSPERO. The registration number is CRD42019122310.

Ethics
No ethical approval was needed to conduct the study

Patient and Public Involvement
Due to the nature of systematic reviews, it was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

Authors' contributions
DB and ER designed and conducted the systematic review. DB wrote the draft of the systematic review and revised it according to the commentaries of ER, SB and TL. DB provided the final version of the manuscript. ER critically reviewed the manuscript and substantially contributed to the final version of the manuscript. SB critically reviewed both the design of the systematic review as well as the manuscript, and was involved in the development of meaningful inclusion criteria. TL critically reviewed the design of the study, made important suggestions for improvement, critically reviewed the manuscript and substantially contributed to the final version of the manuscript. All authors approved the final version of the manuscript.

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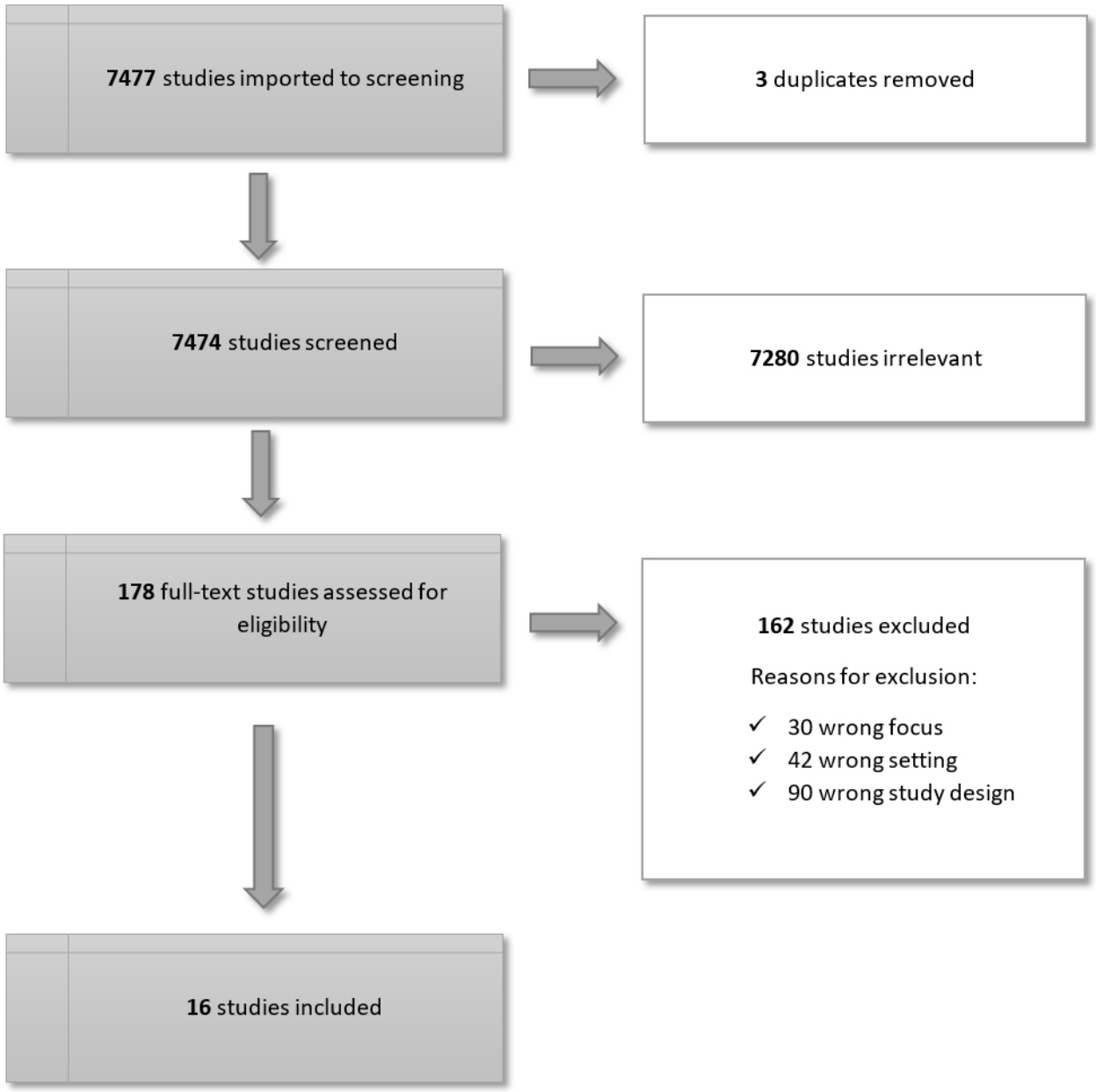
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General Informations				Tool specification							Definitions	
Author s	Year	Title	Journal	Type of tool		Level of measurment	Validation of the tool	Study population	Number of participants (registered and completed)	Year of data collection	Variant defined (CB, CD, CS)	Definition given
Akerjordet, K.; Lode, K.; Severinsson, E.	2012	Clinical nurses' research capacity in a Norwegian university hospital: part 2.	Journal of Nursing Management	**Questionnaire from Australian experience** (Gethinger al 2001) -**59 questions** rated with five point Likert scale + open-ended questions.	**7 AREAS**: exploratory, literature review, design, preparation for action, action, data analysis and writing-up.	Individual	No	nurses	364 out of 600 partecipants (61% response rate).	2011	n/a	Not given
Alison, J. A.;Zafiropoulos, B.;Heard, R.	2017	Key factors influencing allied health research capacity in a large Australian metropolitan health district.	Journal of multidisciplinary healthcare	**Research Capacity in Context Tool (RCCT)** - validated survey .Three domains - contains 18 questions for the organization domain, 19 questions for team domain, and **14 questions for the individual domain**. Rate on a 1–10-point Likert-style scale if the organization, the team, or they as individuals have success or skills in the features described in each item (where 1=no success/skill and 10=high success/skill).	RCCT survey measures research capacity and culture across three domains: organization, team, and individual with the **following AREAS**: 1) skills or success in research at the organization, team (or department), and individual levels; 2) barriers and motivators to performing research; 3) individual research activities completed in the past 12 months; 4) basic information regarding the individual's work role, qualifications obtained (ie, certificate, undergraduate, postgraduate, and PhD), and enrollment in higher degree study (yes/no) or other professional development related to research.	Individual, team and organisation levels.	Yes	Allied health professionals working within the Sydney Local Health District (SLHD	278 out of 514 contacte returned surveys giving a response rate of 54%.	Between November 2015 and May 2016	CB + definition also of research culture and research capacity	Research capacity building describes a process of individual or institutional development which leads to a higher level of skill, with a greater ability to perform research that promotes the health of individuals and communities.
Ekeroma, A. J.; Kenealy, T.; Shulruf, B.; McCowan, L. M.; Hill, A.	2014	Building reproductive health research and audit capacity and activity in the Pacific Islands (BRRACAP) study: methods, rationale and baseline results.	BMC Medical Education	**'Characteristics Questionnaire'** . The questionnaire was developed using a slightly modified tool that was validated in in Australia and made it applicable for the Pacific clinicians. **19 score questions + several opened questions.**	The questionnaire sought demographic professional and personal data that will inform of the participants' **research experience**, their current qualifications, work role and research expectations, and what they perceived to be **barriers** to and enablers of research in their setting. Specific question on **audit skills** are included. The questionnaire was piloted and could be completed comfortably within 30 minutes with responses collated thematically.	Individual + barriers.	The tool was tested using cognitive interviewing of five clinicians	Reproductive Health Clinicians	The 28 participants included 9 nurses/midwives, 17 medical doctors of whom 8 were specialists in reproductive health and 2 other health workers.	From feb 2013 to agust 2014	CB	Not given
Holden, L. ; Pager, S.; Golenko, X.; Ware, R. S.	2012	Validation of the research capacity and culture (RCC) tool: measuring RCC at individual, team and organisation levels.	Australian Journal of Primary Health	**The research capacity and culture (RCC) tool. 15 questions for Individual level (Table 7). Respondents rate on a scale of 1–10.**	Measure indicators of research capacity at individual, team and organisation levels. It **contains a series of statements relevant** to these three domains that **respondents rate on a scale of 1–10**, with one being lowest skill or success level and 10 being highest possible skill or success level. Items are scored separately for each domain. In addition to the scored items described above, **additional questions provide contextual information** to assist in the interpretation of the outcome scores, for example: current work role, professional qualification and resources available to support staff when their role description includes research. Additional questions are also included for needs assessment purposes to explore barriers and enablers to research at both individual and team levels.	Individual, team and organisation levels.	Face validity ofthe RCC tool was assessed by an international expert to ensure all relevant aspects were included and measured appropriately. Face validity was further examined by a group of 12 university staff, both academic and administrative, who were asked to complete the survey and provide feedback. [..] + see methodology since validation was the main aim of the study.	Allied health professional	The RCC tool was administered to a sample of 134 allied health professionals. Approximately 1000 allied health workers were emailed and.	Not specified but before 2010	CB + definition also of research culture	Research capacity building (RCB) is defined as a process of developing sustainable abilities and skills enabling individuals and organisations to perform high quality research (Trostle 1992;D'Auria 2000).
Jule, A.; Furtado, T.; Boggs, L.; Loggerenberg, F. van; Ewing, V.; Vahedi, M.; Launois, P.; Lang, T.	2017	Developing a globally applicable evidence-informed competency framework to support capacity strengthening in clinical research.	BMJ Global Health	**Special Programme for Research and Training in Tropical Diseases (TDR) Global Competency Framework for Clinical Research.**	50 distinct competencies, which can be grouped into **five balanced categories**: Ethics, Quality and Risk Management; Study and Site(s) Management; Research Operations; Scientific Thinking; and Professional Skills.	Individual	In line with Lucia and Lepsinger's recommendations for the validation of competency frameworks,21 the pilot testing will follow	Applicable to all research roles	n/a	n/a	CD	Not given
Matus, J.; Wenke, R.; Hughes, I.; Mickan, S.	2019	Evaluation of the research capacity and culture of allied health professionals in a large regional public health service.	Journal of multidisciplinary healthcare	**Research Capacity and Culture (RCC) tool - Individual (15 questions)** Each question uses a 10-point numeric scale format where 1 is the lowest possible level of skill or success and 10 is the highest possible level of skill or success. There is also an option of "unsure" 15 minutes to complete.	The tool includes **52 questions that examine participants' selfreported success or skill in a range of areas related to research capacity or culture across three domains**, including the Organization (18 questions), Team (19 questions), and **Individual (15 questions).** The RCC tool also includes questions **about perceived barriers to and motivators for undertaking research.** Participants can indicate which, from a list of 18 barriers and 18 motivators, apply to themselves as individuals and within their team and to add any other items that are not included in this list. Finally, the tool includes questions about participants' demographics, work role, professional qualifications, and formal research engagement and what research activities are required and supported in their current work role.	Individual, team and organisation levels	Face validity ofthe RCC tool was assessed by an international expert to ensure all relevant aspects were included and measured appropriately. Face validity was further examined by a group of 12 university staff, both academic and administrative, who were asked to complete the survey and provide feedback. [..] + see methodology since validation was the main aim of the study.	All 875 AHPs and 97 allied health assistants employed by Gold Coast Health (GCH) were invited to participate in a survey consisting of the RCC tool. Students and non-AHPs including medical and nursing staff were excluded.	A total of 369 AHPs accessed the survey during the data collection period. Of them, four did not consent to participate in the survey, 63 provided consent but did not answer any of the questions, 44 incompletely answered the questions, and 258 completed the entire survey. In total, data from 302 AHPs were available for analysis, which represent 30% of the estimated total allied health workforce within our health service.	April' 2017	CB	Capacity building defined as a process of individual and institutional development which leads to higher levels of skills and greater ability to perform useful research (Trostle J. 1992).
Njelesani, J.; Dacombe, R.; Palmer, T.; Smith, H.; Koudou, B.; Bockarie, M.; Bates, I.	2014	A Systematic Approach to Capacity Strengthening of Laboratory Systems for Control of Neglected Tropical Diseases in Ghana, Kenya, Malawi and Sri Lanka.	PLoS Neglected Tropical Diseases	**Questionnaire for laboratory managers**, a **semi-structured interview guide** for use with laboratory stakeholders, a capacity gap checklist for use with the laboratory manager and laboratory staff, and a checklist for ISO 15189 to be used for on-site observations.	n/a	Individual, organisations levels.	We revised the tools after their implementation in each country by conducting a retrospective analysis ofhow the tools contributed or not to the awareness phase in the framework for designing and evaluating a health research capacity-building programme that guided the design of our capacity strengthening tools.	Laboratory scientists, laboratory directors, research staff, WHO staff, ministry representatives, students, human resource and financial staff, donors, and senior academics.	62 semi-structured interviews were conducted, 17 in Malawi, 11 in Ghana, 16 in Kenya, and 18 in Sri Lanka.	2012	CD	Capacity development defined as the ability of individuals, organisations or systems to perform appropriate functions effectively, efficiently and sustainably"(Milen A - 2001)
Petersen, M.;Pain, T.; Fernando, M.	2018	Building Allied Health Research Capacity at a Regional Australian Hospital: A Follow-up Study.	Internet Journal of Allied Health Sciences & Practice	**Cross sectional electronic survey - 14 areas of expertese** responses were set out **as 5-point Likert scales .**	The original survey was modified to capture additional information regarding motivators for participating in research. The data collected included **demographic** variables (age, gender, work-place characteristics), and **level of research experience and support needs**. Participants were also asked to rate the benefits of research and research barriers and enablers. All responses were set out as 5-point Likert scales.	Individual + motivation and barriers .	n/a	Allied health professionals	The response rate for the 2011 survey was 43% (n=248) while the 2015 survey achieved a 37% response rate (n=234 out of 618 AHP in the THHS)	The baseline survey was conducted in 2011 which was compared to a follow up survey in Nov 2014 - March 2015	CB	n/a
Raghavan, R.; Kupfer, L.	2009	Tracking trainees to success.	N/A	**Webbased trainee tracking tool – CareerTrac**	FIC grantees enter both **qualitative and quantitative information about their trainees including:** trainee's name, gender, country of origin, contact information, pre-training and training information, in-training education, accomplishments – such as fellowships, awards, employment, education, publications, funding received, and posters at scientific conferences, Career highlights, leadership, post-training competitive funding, presentations and consultations at scientific meeting. that are related to FIC training. This first of its kind system was developed as a tool for: 1-FIC staff to evaluate the effectiveness and impact of its health research training programs 2- FIC grantees to track their trainees and use the information in progress reports and when applying for other grants.	Individual	n/a	FIC trainees	About 3500 trainees in CareerTrac.	The CareerTrac application has been operational since December 2008.	n/a	n/a
Ried, K.; Farmer, E. A.; Weston, K. M.	2006	Setting directions for capacity building in primary health care: a survey of a research network	BMC Family Practice	**Research spider' tool - 10 areas of reseach with individual level of competency measure with a five point scale ranging from 1** (no experience or interest) **to 5** (high experience or interest).	Part three of the survey sought to ascertain the level of research experience using the visual 'research spider', a simple instrument for self-evaluation of knowledge. The 'research spider' was used to collect information on individual research experience and **interest in research skill development in ten core areas**. These include 'writing a research protocol', using quantitative research methods', 'publishing research', 'finding relevant literature', and 'applying for research funding' [all ten core areas are displayed in Figure 2]. In each area, the level of experience was measured on a five-point scale, from 1 (no experience) to 5 (high experience).	Individual	Smith H, Wright D, Morgan S, Dunleavey J: The 'Research Spider': a simple method of assessing research experience. Primary Health Care Research and Development 2002, 3:139-140.	PRiamry health care . The majority of respondents (55%) worked in allied health and general practice. 'Allied health' included consumer/community health, aged care, child, youth and women's health, chronic illness, mental health, nutrition, occupational therapy, pharmacy. General practice included divisional staff. Other professional areas were nursing, health services including coordinators, managers and information technologists, medical specialists, and academics.	A total of 89 members out ot 229 members (40% of all first year members) participated in the survey. More female health professionals (74%) than male health professionals joined SARNet in 2002/03, and accordingly represented the majority of respondents (76%). The median age of all respondents was 40, ranging from 21 to 62 years. The majority of survey respondents (75%) were residents of metropolitan South Australia, the remainder representing rural South Australia and interstate.	The baseline survey was sent to all members who joined between November 2002 and December 2003.	CB	Capacity buildingdefined as a whole system approach, which promotes participation of new researchers through to more experienced practitioners.
Ried, K.; Farmer, E. A.; Weston, K. M.	2007	Bursaries, writing grants and fellowships: a strategy to develop research capacity in primary health care.	BMC Family Practice	**Semi-structured 40 minute telephone interview** conducted by an external and independent interviewes. **A copy of the research spider was available** to grant holders during the telephone interview. Grant holders were asked to rate their perceived skill level **from 1 = 'no experience' to 5** = 'very experienced' prior to and after their research activity. Grant recipients were **asked to rate the impact of the supported activity on their capacity and confidence to participate in or initiate a research project**, their confidence to seek collegial support for research collaboration, and their general interest in doing future research. Impact was **measured on a 5-point scale,** with a score of 1 reflecting 'no impact', and a score of 5 reflecting	Feedback was sought on acquisition of research skills, publication outcomes, the impact on confidence, and interest in pursuing research in the future in relation to the PHCRED funded project. In total **10 areas** where serached. Figure 3: The 'research spider' was used to assess grant holders' research experience before and after the supported research activity. The level of experience was measured using a five point scale ranging from 1 (no experience) to 5 (very experienced).	Individual + motivation and barriers.	Smith H, Wright D, Morgan S, Dunleavey J: The 'Research Spider': a simple method of assessing research experience. Primary Health Care Research and Development 2002, 3:139-140. 11. Cooke J: A framework to evaluate research capacity building in health care. BMC Fam Pract 2005, 6:44.	Most of the grant holders were allied health professionals (47%). A smaller number were general practitioners (GPs) (21%), Division of General Practice staff (16%), and nursing professionals, medical and primary health care students, and one consumer organisation (Figure 1)	Thirty-four (89%) grant holders participated in the evaluation out of 38 PHCRED funding holders who had completed their projects by early 2006; 21 of 24 bursary holders (88%), 10 of 11 writing grant holders (91%), and 3 of 3 research fellows. The majority of Flinders PHCRED funding recipients lived in metropolitan Adelaide (73.5%), were female (74%), and between 35 and 54 years of age (79%). Nearly 20% of the grant recipients were based in rural South Australia and Victoria, in the Greater Green Triangle (GGT) region.	2006	n/a	n/a
Sarre, G.; Cooke, J.	2009	Developing indicators for measuring Research Capacity Development in primary care organizations: a consensus approach using a nominal group technique.	Health & Social Care in the Community	**Indicators to measure RCD** in institutions. Starting from **Cooke 2005** - Nominal group technique use to improve and test the list.	**9 Principles:** The number of indicators agreed by consensus for eachprincple ranged from 22 indicators for infrastructure to only four indicators each for those relating to research culture and continuity, and sustainability. Pricinples are: infrastructure, linkages and partnerships, dissemination, Skills development, Reseach activity, Close to practice, Leadership, Continuity and sustainability, Research Culture.	Institutional and Individual	No	Stage 1: NHS Trust R&D managers. Stage2: Purposive sample includes nine participants comprised three R&D managers, two research facilitators, two research practitioners (one allied health professional and one general practitioner), one RDSU member with mentoring and capacity-building expertise in primary care and one Primary Care Research Fellow.	Stage 1: over 70 participants who included NHS Trust R&D managers - Stege2: 9 out of 11 invited experts for this study, individuals with experience of research capacity building, doing research, mentoring researchers and those who had been on the receiving end of research capacity-building initiatives.	The study was undertaken between March 2005 and April 2006.	CD	Research Capcity development is 'a process of individual and institutional development which leads to higher levels of skills and greater ability to perform useful research' (Trostle 1992, p. 1321)
Sonstein, S. A.; Namenek Brouwer, R. J.; Gluck, W.; Kolb, H. R.; Aldinger, C.; Bierer, B. E.; Jones, C. T.	2018	Leveling the Joint Task Force Core Competencies for Clinical Research Professionals.	Therapeutic Innovation and Regulatory Science	**Joint Task Force for Clinical Trial Competency (JTF) Core Competecies Framework 2.0**. Use 3 levels of competencies. They are **(1) Fundamental** ("Can perform the task/and or exhibit the knowledge at an essential or fundamental level; may require some coaching or supervision"), **(2) Skilled** ("Can perform task or skill independently, consistently, accurately, and has a moderate level of expertise. Efficient and high-quality work; able to independently navigate resources and uses tools well"), and **(3) Advanced** ("Demonstrates advanced skills and knowledge and the ability to teach, coach, or supervise others. Consistently applies critical thinking and problem solving".	The JTF Framework is composed of **8 Domains (Figure 1) 1- Scientific Concepts and Research Design 2- Ethical and Participant Safety Considerations 3- Investigational Products Development and Regulation 4- Study Operations (Good Clinical Practice) 5- Study and Site Management 6 Data Management and Informatics 7-Leadership and Professionalism 8-Communications and Teamwork- and 47 core competency statements** that were aligned and harmonized utilizing published statements of core competency requirements to encompass the entire clinical research enterprise.	Individual	Ultimately, the profession of clinical research was recognized by the Commission on Accreditation of Allied Health Education Programs (CAAHEP); academic programs in clinical research are now eligible for CAAHEP accreditation based on JTF core competency-informed educational standards.	Applicable to all research roles	n/a	2016	CB	n/a

Figure 2 - *PRISMA screening diagram*



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3/4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4/16
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6/7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7/8
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	8
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	10
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	11
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	9
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	n/a
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	14
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/a
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10/20/21
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	14
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	14
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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Measuring Research Capacity Development In Healthcare Workers: a systematic review.

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ABSTRACT

Title: Measuring Research Capacity Development In Healthcare Workers: a systematic review.

Objectives: A key barrier in supporting health research capacity development (HRCD) is the lack of empirical measurement of competencies to assess skills and identify gaps in research activities. An effective tool to measure HRCD would help inform teams to undertake more locally-led research. The objective of this systematic review is to identify tools measuring Healthcare Workers' (HCW) individual capacities to conduct research.

Design: Systematic review and meta-analysis using PRISMA checklist for reporting systematic reviews and meta-analyses and CASP checklist on critical appraisal for qualitative studies.

Data sources: 11 databases were searched from inception to 16 January 2020. The first 10 pages of Google Scholar results were also screened.

Eligibility Criteria: We included papers describing the use of tools/ to measure/assess HRCD at an individual level among healthcare workers involved in research. Qualitative, mixed and quantitative methods were all eligible. Search was limited to English language only.

Data extraction and synthesis: Two authors independently screened and reviewed studies using Covidence, and performed quality assessments using the extraction log validated against the CASP qualitative checklist. The content method was used to define a meta-narrative analysis.

Results: The titles and abstracts for 7,474 unique records were screened and the full text of 178 references were reviewed. 16 papers were selected: seven quantitative studies; one qualitative study; and five mixed-method studies. Tools with different levels of accuracy in measuring HRCD in HCW at the individual level were described. The Research Capacity and Culture Tool and the "Research Spider" tool were the most commonly defined. Other tools designed for ad hoc interventions with good generalizability potential were identified. Three papers described health research core competency frameworks. All tools measured HRCD in HCW at an individual level with the majority adding a measurement at the team/organisational level, or data about perceived barriers and motivators for conducting health research.

Conclusions: Capacity building is commonly identified with pre-post intervention evaluations without using a specific tool. This shows the need for a clear distinction between measuring the outcomes of training activities in a team/organisation, and effective actions promoting HRCD. This review highlights the lack of globally applicable comprehensive tools to provide comparable, standardised and consistent measurements of research competencies.

Review registration and funding: PROSPERO (CRD42019122310). Independent research.

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ARTICLE SUMMARY

Strengths

- Thoroughly-conducted systematic review collecting data from all major existing databases and grey literature.
- Topic not previously addressed in other reviews searching for tools to measure health research capacity building at individual level.
- Brief overview of the identified tools to measure health research capacity building at individual level highlighting strengths and weaknesses of them.

Limitations

- Complex identification of relevant studies due to the lack of clarity on a common definition and terminology to identify HRCD.
- None of the studies use the standard reporting procedures for qualitative or quantitative research.

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INTRODUCTION

In 2004 the Global Forum for Health Research highlighted the challenge for low and middle-income countries (LMICs) to have the capacity to perform effective and locally-led health research which address the major health problems affecting their own populations (1-3). Twenty years later, LMICs still carry 90% of the global disease burden, but only 10% of global funding for health research is devoted to addressing these persistent health challenges (4). Health research capacity development (HRCD) for healthcare workers (HCW) has been recognised as a critical element to overcoming global health challenges, especially in LMICs (5). For too long HRCD in LMICs has been documented through training programmes which enable local teams to participate in externally sponsored trials, creating a false appearance of growth and generating dependence on foreign support (6, 7).

The process of progressive empowerment is usually referred to as capacity development (8). This term has been used in multiple areas and applied in different sectors to develop new or existing competencies, skills, strategies at a macro or individual level (9). In the field of health, research capacity development should support HCW in generating local evidence-based results to inform policy and improve population health. The three health-related Millennium Development Goals, and more recently the targets "B" and "C" of the Sustainable Development Goals, all support the adoption of new strategies to strengthen the capacity of HCW in all countries in performing their job and engaging in research (10-12). One of the critical barriers in supporting HRCD is the lack of empirical measurement of competencies in relation to the performance of research activities. Existing frameworks and tools have been developed for a particular purpose in a particular context (13) (14). Others have identified barriers HCW encounter in engaging in research or have monitored and evaluated targeted training activities (15). This systematic review aims to identify tools to measure individual HCW's capacities to conduct research.

METHODS

The PRISMA checklist (16) for reporting systematic reviews and meta-analyses and CASP checklist (17) on critical appraisal for qualitative studies were used to design this systematic review and to refine the extraction log according to recognised guidelines.

Inclusion and exclusion criteria

The aim of the systematic review was to identify existing tools which measure individual capacities in conducting research in HCW. The inclusion and exclusion criteria were defined in advance and documented using an adapted version of a SPIDER table (Table 1). The primary population of interest were all health-related professionals or HCW involved in research activities. HCW delivering health services when research was not considered as the focus of the study were excluded. Occupational health research was excluded. Studies about volunteers, defined as people offering their services to support health activities with no specific training as health professionals, were also excluded. Initially, only HCW working in LMICs were included, but this limitation was removed to identify any tool measuring HRCD in any setting. The Phenomenon of Interest was defined as: assessing HRCD; or identifying tools, frameworks and templates designed to assess HRCD. A comprehensive range of terms including synonyms for "assess", "tool" or "development" was used. Studies were excluded which mentioned components that could be considered to assess, measure and 'give evidence to' research capacity development, but were not presented in any capacity development context. In addition, since the concept of capacity development is widely applied to different settings, studies on areas unrelated to health, such as "air pollution", "financial capacity" or "tobacco", were also excluded. The study design criteria were broad, to include qualitative, quantitative and mixed methods papers. Further criteria of eligibility included in the SPIDER table refers to the quality of the study (Evaluation) and the Research type.

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Table 1 - ***SPIDER diagram - Inclusion and exclusion criteria***

SPIDER	*Inclusion*	Exclusion
Sample (population involved)	Healthcare workers (HCW) involved in research defined as: Health professionals involved in research activities at every step of research and with any health-related professional profile working in health centres.	Heath workers delivering health services when research is not considered People involved in occupational health research Volunteers defined as people offering their services to support health activities with no specific training as health professionals.
Phenomenon of Interest	Broadly, assessing health research capacity development. Specifically, analysing tools, frameworks, templates to assess and measure health research capacity development/building/strengthening. Key words: Tool – framework - assess – measure – give evidence - capacity development – capacity building – capacity strengthening - competency improvement – professional development - mentoring	Include specific components that could be considered as aiming to assess, measure and give evidence to research capacity development, but are not presented in any capacity development context. Non-health-related capacity development research such as “air pollution”, “financial capacity”, “tobacco” or other areas unrelated to health research emerging from the search refining.
Design (nature of the intervention described)	Qualitative literature including interview, focus groups, surveys, grounded theory, ethnography, interpretative phenomenological analysis, content analysis, thematic analysis. Quantitative literature including score systems, competencies measurement. Mixed methods papers combining qualitative and quantitative approach	Studies that do not meet the inclusion criteria for design.
Evaluation	Acceptable: meeting high and moderate score in quantitative assessment tool EPHPP and measure qualitative validity with COREQ evaluation method if applicable.	Studies that do not meet the inclusion criteria for evaluation.
Research Type	Qualitative, mix and quantitative methods	Do not have English language abstract

Information sources and search strategy

Eleven databases were searched from inception to 16 January 2020: Ovid Medline; Ovid Embase; Ovid PsycINFO; Ovid Global Health; Ebsco CINAHL; ProQuest Applied Social Sciences Index & Abstracts (ASSIA); ProQuest Sociological Abstracts; ProQuest Dissertations & Theses Global; Scopus; Web of Science Core Collection; and the WHO Global Index Medicus Regional Libraries. The first ten pages of results from Google Scholar were also screened. The search strategies used free-text terms and combinations of the relevant thesaurus terms, limited to English language publications only, to combine terms for capacity building, measuring, and health research. The“NOT” command was used to exclude papers about students, postgraduates, tobacco, air pollution, and a variety of other concepts to minimise the number of irrelevant results (Box 1 for a full set of search strategies).

BOX 1:

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Database: Medline (Ovid MEDLINE® Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE® Daily and Ovid MEDLINE®) 1946 to present

Search Strategy:

---

1 Capacity Building/ (1965)
2 (capacit* adj2 build*).ti,ab. (5789)
3 (capacit* adj2 develop*).ti,ab. (3591)
4 (capacit* adj2 strengthen*).ti,ab. (924)
5 (competenc* adj2 improv*).ti,ab. (1460)
6 ((professional* adj2 develop*) and (competenc* or capacit*)).ti,ab. (1747)
7 1 or 2 or 3 or 4 or 5 or 6 (13649)
8 Mentoring/ (820)
9 mentor*.ti,ab. (13369)
10 (assess* or measur* or evaluat* or analys* or tool* or equip*).ti,ab. (9653076)
11 "giv* evidence".ti,ab. (3814)
12 framework*.ti,ab. (231138)
13 8 or 9 or 10 or 11 or 12 (9763562)
14 Research/ (196782)
15 clinical.ti,ab. (3158817)
16 (health* and research*).ti,ab. (337604)
17 14 or 15 or 16 (3588891)
18 7 and 13 and 17 (3433)
19 18 (3433)
20 limit 19 to english language (3346)
21 (student* or graduate or graduates or postgraduate* or "post graduate*" or volunteer* or communit* or tobacco or "climate change" or "air pollution" or occupational or "financial capacity" or informatics or "IT system" or "information system" or transport or "cultural competenc*" or disabili* or trauma).ti,ab. (1828113)
22 20 not 21 (1673)

Google Scholar – Screen the first 10 pages of results

Sorted by relevance:

("capacit* build*"|"build* capacit*"|"capacit* develop*"|"develop* capacit*"|"capacit* strengthen*"|"strengthen* capacit*"|"professional* develop*"|"completenc* improv*"|"improv* competenc*")("health* research*"|clinical)

https://scholar.google.co.uk/scholar?q=(%22capacit*+build*%22%7C%22build*+capacit*%22%7C%22capacit*+develop*%22%7C%22develop*+capacit*%22%7C%22capacit*+strengthen*%22%7C%22strengthen*+capacit*%22%7C%22professional*+develop*%22%7C%22completenc*+improv*%22%7C%22improv*+competenc*%22)(%22health*+research*%22%7Cclinical)&hl=en&as_sdt=0,5

Study selection

Two researchers, DB and ER independently screened and reviewed studies using the Covidence online tool for systematic review (18). In case of disagreement, DB and ER discussed the abstracts in question. After consensus on inclusion was reached, the full-texts of all included studies were rechecked for inclusion by DB and confirmed by ER.

Study analysis procedure

Data from selected papers were extracted, and quality assessments performed using an extraction log created and validated against the CASP checklist (17) on critical appraisal for qualitative studies. Macro areas of interest in the log were: general information on the paper such as author and title, main focus and study design. The source of funding, conflict of interests and ethics approval were also recorded. A separate section of the extraction log recorded the characteristics of the tool used or described in each selected paper (Figure 1). The extraction log also included specific sections considering the study design, the methodology and the main findings of each paper. Furthermore, a dedicated section of the log collected data on the quality of each study, analysing selection biases and a critical appraisal

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derived from the CASP checklist. If a definition of Capacity Development was given, the definition was collected. Some of these sections of the extraction log are not present in Figure 1 since it focuses on the description of the identified tool.The content method was used to define a meta-narrative, described in the Discussion section.

RESULTS

Database search and results screening

In December 2018, the first round of the search was performed in 11 different databases and in Google Scholar using the search strategy described in box 1. A total of 13,264 suitable records were found. A total of 6,905 duplicates were removed, resulting in 6,359 unique records for inclusion screening by title and abstract (Table 2), which was performed throughout 2019. In January 2020 an additional search for papers published or included in publication databases in 2019 was performed using the same search strategy and resulted in 15,775 papers and after removal of duplications, a total of 1,118 papers were found. These papers were then added to the 6,359 papers identified from the first search. 7,474 unique papers were included for title and abstract inclusion screening (three duplicate records were removed in Covidence).

Table 2 - *Search Results*

	Search dates: 20-21/12/18	Search date: 16/01/2020
Ovid Medline (searched on 20/12/18)	1673	1937
Ovid Embase (searched on 20/12/18)	2344	2721
Ovid PsycINFO (searched on 20/12/18)	597	619
Ovid Global Health (searched on 20/12/18)	566	676
EBSCO CINAHL (searched on 20/12/18)	1376	1663
ProQuest Sociological Abstracts (searched on 20/12/18)	305	371
ProQuest ASSIA(searched on 20/12/18)	1463	2036
ProQuest Dissertations & Theses Global (searched on 20/12/18)	279	312
Scopus (searched on 21/12/18)	2230	2575
Web of Science – Core Collection (searched on 21/12/18)	1900	2195
WHO Global Index Medicus (searched on 21/12/18)	531	670
Total	13264	15775
Total after deduplication	6359	7469
16/01/2020: Unique to databases since 20-21/12/18		1118
TOTAL TO SCREEN		7477

The 7,474 unique relevant studies identified were uploaded to the Covidence online tool. Two researchers, DB and ER independently screened the studies including or excluding studies according to the criteria in the SPIDER table (Table 1). The Covidence system presented the divergent opinions between the two researchers and disagreements were resolved with a dedicated discussion session. A total 7280 studies were considered irrelevant. The full text papers for the remaining 178 references were reviewed. Reasons for exclusion were identified by streamlining the SPIDER table criteria into three main criteria: wrong setting, irrelevant study design and wrong focus of the study. A reason for exclusion was assigned to each paper. All 178 studies described some form of activity to measure the competencies related to performing health research. Thirty were excluded because they were literature reviews on a different aspect of health research or because they described a general perspective on the topic of health capacity development without offering any specific measurement or without reference to research. In addition, 42 studies were excluded because of the wrong setting, since competencies were measured at the level of research institutions or within a specific network.

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An additional 90 studies were excluded because the study design did not match the inclusion criteria: 38 studies described the use of a measurement tool tailored to the context (e.g. specific profession, intervention or setting) and not at the individual level; the remaining 34 studies were excluded because there was no mention of a specific tool to measure HRCD. The final 18 papers reported the use of an evaluation tool, but the tool was an *ad hoc* pre-post intervention questionnaire with low potential of applicability in a context different from the one described in the paper. A total of 162 studies were therefore excluded, leaving 16 studies for this review (Figure 2).

Analysis of the findings across the selected papers

A total of 16 studies met the inclusion criteria set for this systematic review (19-34). The 16 articles were analysed using the extraction log created and validated against the CASP qualitative checklist.

The results are summarised in Table 2. None of the papers were published before 2006 and only 9 of them were published after 2014 (20, 21, 23-26, 31, 33, 34). The majority (n = 13) applied a tool in high-income settings (19, 20, 22-24, 26-32, 34). Seven papers described the use of tools in Australia (20, 22, 24, 26, 28, 29, 34), three in low and middle-income countries (one in Ghana, Kenya, Malawi, Sri Lanka (25), one in the Pacific Islands (21) and one in the Philippines (33)), one in Europe (Norway) (19), one in the US (32), and one measured HRCD in a group linked to a specific intervention located in multiple areas of the world. Three of them described the creation of a tool without applying it to any specific context (23, 30, 31), but they were all designed by research groups in high-income countries (one in the US and two in the UK).

All of the selected studies applied quantitative, qualitative and mixed-method analyses. The preferred approach (n=7) was to generate quantitative data using an HRCD tool (20, 24, 26-28, 33, 34). One-third of the studies (n = 5) used a mixed methods approach (19, 21, 22, 25, 29); quantitative tools were associated with semi-structured interviews, or in some cases qualitative questions were added to the questionnaire. The three studies describing the creation of a tool were not analysed under this methodological category.

Of the 16 selected studies, three used the term “capacity development” (23, 25, 30), and two included a definition of the concept (25, 30). Seven papers used “capacity building” (20-22, 24, 26, 28, 31), of which four also included a definition (20, 22, 24, 28). In two papers, the capacity building definition was associated with the definition of “research culture” (20, 22). Two additional papers used alternative generic terms like “research capacity” (33) or “research self-efficacy” (32). Four papers did not refer to any specific term and therefore no definition was given (19, 27, 29, 34).

Five of the 16 selected papers openly declared no conflict of interests (22-24, 28, 31). Eight stated the source of funding used to carry out the activities described (19, 21, 23, 27-31). The number of participants in the studies varied from 28 enrolled participants for a qualitative study (21) to 3500 users of an online measurement tool (27).

Analysis of the tools from the selected papers

The tools described or used in the 16 selected papers varied in nature, length and applicability. In general, even when there were similarities, each paper described a different perspective on the use of a tool. Four papers applied a questionnaire-type tool to assess research competencies and skills (19, 21, 25, 33). The length of these questionnaires varied from 19 (21) to 59 (19) health research capacity

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related questions, with the addition of open-ended qualitative questions in two studies (19, 21), and a structured interview in another study (25).

Three studies (22, 24, 34) used, with a range of adaptations, the Research Capacity and Culture Tool (RCC) and one study (20) revised the RCC tool into a Research Capacity and Context Tool (RCCT) referencing RCC as a primary source. Another recurrence in the papers was the use of the "Research Spider" tool (28, 29). Again, the original tool had been adapted to the context, and in one case (29), the tool was used as a base for qualitative research on HRCD. Two additional papers described tools designed *ad hoc* to measure the impact of an intervention (CareerTrac (27) and Cross-Sectional Electronic Survey (26)). These last two papers were not excluded under pre-post intervention since the action was wider, at a programme level, and the tool used to measure HRCD was the main focus of the paper. Furthermore, another paper described a tool for a specific category of HCW (Nursing Research Self-Efficacy Scale -NURSES) (32). Three papers (23, 30, 31) focussed on the creation of a new tool and described the process of identifying a set of competencies required to run health research. The outcome of two of them was defined as a "core competency framework" (23, 31). The third defined the outcome of the analysis as a "set of indicators" (30).

In terms of the target population, the identified tools aimed to measure HRCD in a range of different HCW professions. One-third of the papers (n=5) focused on measuring HRCD on Allied Health Professionals (AHPs) (20, 22, 24, 26, 34). Nurses were the main focus in two other studies (19, 32), and four studies applied a tool to a range of health professions (ranging from laboratory scientists to data managers) (21, 25, 28, 29). Two other papers focused on groups linked to a specific intervention (27, 33). All 16 papers included, alongside HCW, representatives of technical professions in health such as managers, directors, faculty members, consumer organisation representatives. In the case of the three papers describing the creation of a new tool, they suggest that these tools would be applicable to all research roles (23, 30, 31).

As per inclusion criteria, the main level of measurement of the tools was at the individual level. Seven papers only measured HRCD at the individual level (19, 23, 28, 29, 31-33). Three papers added to the individual level of measurement by including information on the perceived barriers in performing health research (21, 26, 29); of these three, two also focused on understanding what motivates HCW to become involved in health research (26, 29). The five studies, which used the RCC tool and its variants, included the measurement of HRCD not only at the individual level, but also at the team and organisational level (20, 22, 24, 25, 34). One paper described the creation of a tool designed to be used at the organisational level, but embedded a measurement of HRCD at the individual level as well (30).

The most common way a selected tool was validated was by referencing the main paper that described the selected tool and its validation process (n = 6) (23, 28, 29, 32-34). This was the case for some of the *ad hoc* questionnaires (23, 33, 34), of the "Research Spider" tool (28, 29), and of the NURSES tool (32). Papers which described an original process or used modified versions of an original tool validated the tool through a contextual validation process described in the paper (21, 22, 24, 25, 31). These validation processes included a consultation of a panel of experts (22, 24, 31) or a reiterative process of validity (21, 25). One paper stated that the tool used was a validated tool without referencing the process or tool (20).

Overall, only two papers (23, 31) focussed specifically on tools to measure HRCD on a wider level, without linking the measurement to a specific group or a geographical area which was done in the majority of papers (19, 24, 25, 28, 29, 33). In four cases the tools described were adapted to identify

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determinants or barriers of HRCD in a defined setting (20, 30, 34) or to promote HRCD in relation to a specific disease or research topic (21). In other cases, the papers focussed on a tool aiming to assess the impact of specific interventions or programmes on HRCD (26, 27).

DISCUSSION

Summary of evidence

This systematic review aimed to identify tools which measure individual capacities in conducting research in HCW; the 16 included articles (19-34) which demonstrated that tools to measure HRCD in HCW are available, even if they are limited in number. In most cases, the identified tools do not originate from the need to measure and foster HRCD as a necessary strategy to promote research capacity. There is, therefore, a need to design more comprehensive tools which are globally applicable and able to provide comparable, standardised and consistent measurements of research competencies.

The importance of measuring HRCD has only been recognised recently (15). From the date of publication of the identified papers show, the contribution that health research can offer in capacity development at a personal level only begun in the first decade of this new millennium. Almost half of the selected papers (n= 7) refer to studies whose data have been collected after 2014 (20, 21, 24, 26, 31, 33, 34). Of note is the high number of new publications which were retrieved from the academic databases (1,118 papers) when the search strategy was re-run in 2020.

The questionnaire was the most commonly used method for assessing research skills and competencies. Almost two-thirds of the papers (n = 10) (19, 20, 22, 24, 26, 28, 29, 32-34) based the measuring system of different research skills at a personal level using a 5 point Likert scale (n = 6) (19, 26, 28, 29, 32, 33) or a 10 point scale (n = 4) (20, 22, 24, 34). This choice highlights the need for a validated quantitative tool based on a set of competency related questions that can bring standardisation, comparability and consistency across different roles and contexts. However, the extensive use of mixed methods, combining quantitative questionnaires with other qualitative instruments, reflects that HRCD depends on a complex series of components that need to be identified both qualitatively and quantitatively.

By not limiting the selection of articles for this review to those tools used in low and middle-income countries (LMICs), this review has revealed that most of the tools identified were used in high-income settings. It is important to note that excluding pre/post-intervention assessments significantly reduced the inclusion of studies performed in LMICs. This finding highlights that although health systems in LMICs may benefit from providing evidence for HRCD (5), they are rarely the focus of the HRCD literature. Most of the measurements of HRCD in lower-income settings appears, in fact, to be narrowly linked to the measurement of the effectiveness of training offered for a specific study or limited to a particular disease. Even when the perspective is broader than a particular study, it is mostly limited to the evaluation and sustainability of training programmes and not linked to a plan of career progression and research competency acquisition. More attention should therefore be given in creating tools which are able to measure, support and promote long-lasting research capabilities in the perspective of professional growth for HCW.

Three essential findings of this systematic review support a change in the perception of HRCD and the tools needed to measure it. Firstly, many of the excluded papers (42 out of 162 excluded papers from the last round of analysis) focused exclusively at the institutional level of measuring research capacity. This is mostly because training interventions are designed to prepare a team to run a study and rarely

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to promote individual HRCD (1, 35, 36). In some cases, the measurement via a tool is also an exercise to demonstrate the investment in training activities for reporting purposes (37, 38). It is therefore important to start promoting a more effective research culture which is independent from specific diseases or roles. This progression could be achieved by championing systems which measure the changes in research capacities at a team and personal level using a globally applicable tool. Most of the tools excluded were evaluation tools designed for, or used in a specific setting and thus not suitable for a comparable, standardised and consistent analyses of long term research competency acquisition strategies.

Secondly, papers that focused on measuring HRCD at the individual level confirmed that research is seen as an opportunity to learn the cross-cutting skills needed in healthcare. A defined set of standardised competencies required to conduct research could be used to measure an individual, team and organisation's abilities. This was the focus of two papers (23, 31) which identified a framework of core competencies. Most of the tools (n = 7) were designed to be applied to a wider variety of health professions (21, 23, 25, 28-31). HRCD can be accessed at different entry points depending on the specific job title, but the set of skills acquired is common and shared among the research team (1). The approach on assessing these inter-related competencies should therefore be global and not role or disease-based (39). The measurement at an individual level is essential to promote a consistent and coherent career progression for each person and role (40). However, the overall capability in running research programmes should be measured at a team level where all roles and competencies complement each other, skills are made visible, and measurable as a whole against an overall competency framework. Individual and institutional/team levels are therefore, two aspects of HRCD that grow together supported by a common comparable, standardised and consistent tool.

Thirdly, the lack of a standard definition for HRCD can lead to post-training evaluations being categorised as HRCD activities. Although pre/post training evaluations are important, it might be helpful to define what a 'structured action' is to promote HRCD. As previously mentioned, the term "capacity development" is not universally used, with many synonyms such as "research capacity" or "capacity strengthening", creating the possibility of different interpretations. Furthermore, inconsistent terminology was found in describing activities in support of HRCD that in reality, were very similar (e.g. workshop, training, course). Steiner *et al.* (41) suggest that there should be a standard definition in the context of educational capacity development. This suggestion, alongside a common taxonomy to describe health professions, would support the identification of HRCD as a defined process with specific characteristics and not with a general effort for research training.

The most common tool identified in this review was the Research Capacity and Culture Tool (RCC) (20, 22, 24, 34). The RCC consists of 52 questions that examine participants' self-reported success or skill in a range of areas related to research capacity or culture across three domains including the organization (18 questions), team (19 questions), and individual (15 questions). The RCC tool includes questions on perceived barriers and motivators for undertaking research. The respondents of the RCC are asked to rate a series of statements relevant to these three domains on a scale of 1–10, with one being lowest and ten being the highest possible skill or success level. It represents a good example of a comprehensive tool. As confirmed by the review findings, a potential limitation is its application mainly in an Australian context and almost exclusively to measure HRCD in AHPs (22, 24, 34). The generalizability of the tool should thus be confirmed. Nevertheless, the RCC represents a strong example of how having a tool refined around a context, and a specific health profession can be an incentive in measuring HRCD.

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Another tool highlighted by this review was the "Research Spider" tool (28, 29, 42). This tool collects information on individual research experience and interest in research skill development in ten core areas. These include "writing a research protocol", "using quantitative research methods", "publishing research", "finding relevant literature", and "applying for research funding". In each area, the level of experience is measured on a five-point Likert scale, from 1 (no experience) to 5 (high experience). The primary aim of the "Research Spider" is to be a flexible tool. This flexibility is confirmed in two studies (28, 29) which used the "Research Spider", with one (28) using it as the main measurement, and the other (29) as a quantitative base for qualitative semi-structured interviews. The advantage of this tool is that it provides a visual overview of personal research competencies. However, although the limited number of measurement areas (n = 10) makes the tool a good initial evaluation instrument, it does not offer a specification of the sub-skills of each area.

A critical mention should be reserved for the two papers which described the creation of a comprehensive research core competency framework (23, 31). Despite no specific tool being described and the competency scores being visualised by using a spider diagram, these studies present the most accurate overview of the skills required in running research programmes related to health. As mentioned before, a tool which applies a scoring system to the list of competencies identified by these frameworks has the potential of being widely applicable and reliable. This wide applicability and the absence of explicit biases in measuring research skills improvement, can foster a more robust approach to research in health. The measurement of HRCD unrelated to specific interventions would maximise the benefit of research at every level. At a personal level, it would clarify a potential career progression path highlighting possible gaps; at the team level, it would support a multidisciplinary approach to health challenges; and at an institutional level, the measurement of HRCD would make the know-how generated by the international scientific community accessible to a broader group of local health workers. Overall, health practice at a global scale would benefit from the incentive of getting involved in research derived from measuring the impact of it on improving competencies. Thus positive outcomes of measuring HRCD could place the issue of universal transferability, and applicability of research methodology and results at a higher level of priority in the design of health research projects.

Limitations of the Systematic Review

Methodological limitations are recognised for this systematic review. Firstly, there is a lack of clarity on a common definition and terminology to identify HRCD which complicates the search strategy. A long reiteration process was necessary when developing the search strategies for the databases to try and include all the possible variants used to define "tool", "capacities" and "development". Despite this effort, some studies may have been missed. Secondly, there was a lack of studies which referenced a standard reporting procedure, despite the presence of standards available for reporting qualitative or quantitative research (43-45) as well as for mixed-method research (46). Other limitations typical for reviews may also apply. Finally, while this review has attempted to be as comprehensive as possible, some sources might not have been detected due to the challenge in finding all the relevant grey literature, and the restriction to English language sources only.

CONCLUSIONS

Sixteen studies using or describing tools to measure HRCD were identified and analysed in this systematic review (19-34). Identifying capacity development with pre-post intervention evaluations or to generically evaluate capacity development without using a tool was common. Furthermore, the majority of papers were not included due to measuring HRCD at an institutional level and not at an

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individual level. This finding highlights the need for a clear distinction between simply measuring training activity outcomes in HCW and effective action promoting HRCD for HCW. Further research on harmonisation and clarification of terms related to HRCD would contribute to solving this issue.

Identified tools were analysed on the basis of their frequency within the selected papers. The most recurrent tool described was the Research Capacity and Culture Tool (RCC) based on a 1 to 10 scoring system in 3 areas of research linked to institutional, team and individual level (20, 22, 24, 34). The second most recurrent tool was the "Research Spider" tool which assesses ten core areas of research scored on a 1 to 5 Likert scale (28, 29). A variety of other tools, mostly questionnaire-based, were identified, and in most cases, a broader applicability than described in the specific context of the paper may be possible. Two frameworks systematising research core competencies were identified (23, 31). The potential of tools derived from these frameworks could be significant. The competency areas in conducting research identified by the two frameworks mostly coincide, but the depth of the evaluation in each area varied in each tool. The applicability of each tool depends, therefore, on the context and on the level of accuracy needed

Future directions for HRCD include the design of a standardised, comparable and consistent tool to measure individual HRCD not linked to training evaluation, but support a long term research competencies acquisition strategy. In addition, the harmonisation of definitions and terminologies used in identifying HRCD actions and processes could facilitate standardisation and comparability of HRCD strategies.

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Data sharing:
All data relevant to the study are included in the article. Data are available upon reasonable request.

Competing interests
The authors declare that they have no competing interests.

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Review registration
The protocol for this systematic review has been registered with PROSPERO. The registration number is CRD42019122310.

Ethics
No ethical approval was needed to conduct the study

Patient and Public Involvement
Due to the nature of systematic reviews, it was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

Authors' contributions
DB and ER designed and conducted the systematic review. DB wrote the draft of the systematic review and revised it according to the commentaries of ER, SB and TL. DB provided the final version of the manuscript. ER critically reviewed the manuscript and substantially contributed to the final version of the manuscript. SB critically reviewed both the design of the systematic review as well as the manuscript, and was involved in the development of meaningful inclusion criteria. TL critically reviewed the design of the study, made important suggestions for improvement, critically reviewed the manuscript and substantially contributed to the final version of the manuscript. All authors approved the final version of the manuscript.

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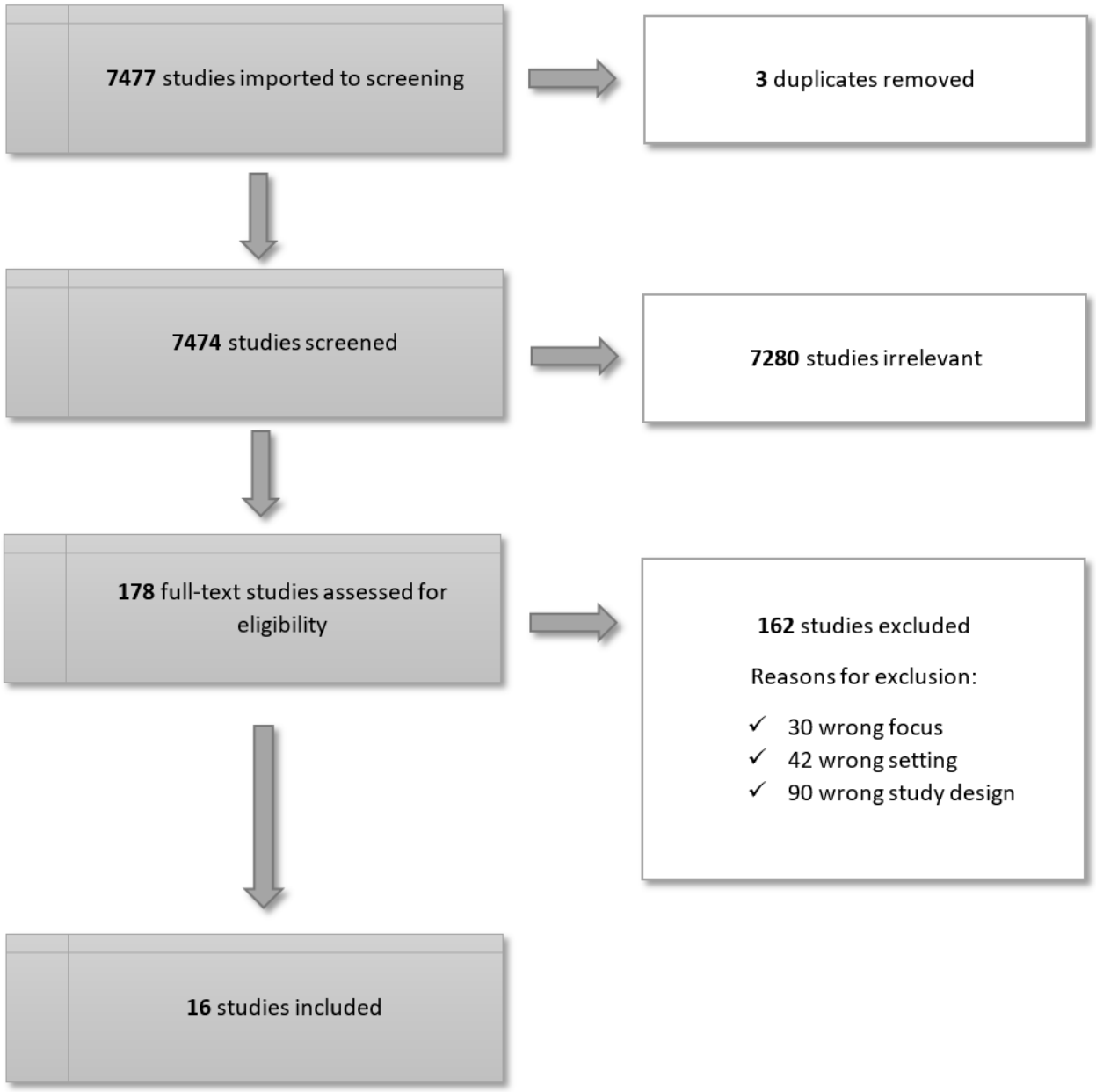
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Figure 1 - Extraction log

General Informations				Tool specification							Definitions	
Author s	Year	Title	Journal	Type of tool		Level of measurment	Validation of the tool	Study population	Number of participants (registered and completed)	Year of data collection	Variant defined (CB, CD, CS)	Definition given
Akerjordet, K.; Lode, K.; Severinsson, E.	2012	Clinical nurses' research capacity in a Norwegian university hospital: part 2.	Journal of Nursing Management	**Questionnaire from Australian experience** (Gethinget al 2001) -**59 questions** rated with five point Likert scale + open-ended questions .		Individual	No	nurses	364 out of 600 partecipants (61% response rate).	2011	n/a	Not given
Alison, J. A.;Zafiropoulos, B.;Heard, R.	2017	Key factors influencing allied health research capacity in a large Australian metropolitan health district.	Journal of multidisciplinary healthcare	**Research Capacity in Context Tool (RCCT)** - validated survey .Three domains - contains 18 questions for the organization domain, 19 questions for team domain, and **14 questions for the individual domain**. Rate on a 1–10-point Likert-style scale if the organization, the team, or they as individuals have success or skills in the features described in each item (where 1=no success/skill and 10=high success/skill).	RCCT survey measures research capacity and culture across three domains: organization, team, and individual with the **following AREAS**: 1) skills or success in research at the organization, team (or department), and individual levels; 2) barriers and motivators to performing research; 3) individual research activities completed in the past 12 months; 4) basic information regarding the individual's work role, qualifications obtained (ie, certificate, undergraduate, postgraduate, and PhD), and enrollment in higher degree study (yes/no) or other professional development related to research.	Individual, team and organisation levels.	Yes	Allied health professionals working within the Sydney Local Health District (SLHD	278 out of 514 contacte returned surveys giving a response rate of 54%.	Between November 2015 and May 2016	CB + definition also of research culture and research capacity	Research capacity building describes a process of individual or institutional development which leads to a higher level of skill, with a greater ability to perform research that promotes the health of individuals and communities.
Ekeroma, A. J.; Kenealy, T.; Shulruf, B.; McCowan, L. M.; Hill, A.	2014	Building reproductive health research and audit capacity and activity in the Pacific Islands (BRRACAP) study: methods, rationale and baseline results.	BMC Medical Education	**'Characteristics Questionnaire'** . The questionnaire was developed using a slightly modified tool that was validated in in Australia and made it applicable for the Pacific clinicans. **19 score questions + several opened questions.**	The questionnaire sought demographic professional and personal data that will inform of the participants' **research experience**, their current qualifications, work role and research expectations, and what they perceived to be **barriers** to and enablers of research in their setting. Specific question on **audit skills** are included. The questionnaire was piloted and could be completed comfortably within 30 minutes with responses collated thematically.	Individual + barriers.	The tool was tested using cognitive interviewing of five clinicians	Reproductive Health Clinicians	The 28 participants included 9 nurses/midwives, 17 medical doctors of whom 8 were specialists in reproductive health and 2 other health workers.	From feb 2013 to agust 2014	CB	Not given
Holden, L. ; Pager, S.; Golenko, X.; Ware, R. S.	2012	Validation of the research capacity and culture (RCC) tool: measuring RCC at individual, team and organisation levels.	Australian Journal of Primary Health	**The research capacity and culture (RCC) tool. 15 questions for Individal level (Table 7). Respondents rate on a scale of 1–10.**	Measure indicators of research capacity at individual, team and organisation levels. It **contains a series of statements relevant** to these three domains that **respondents rate on a scale of 1–10**, with one being lowest skill or success level and 10 being highest possible skill or success level. Items are scored separately for each domain. In addition to the scored items described above, **additional questions provide contextual information** to assist in the interpretation of the outcome scores, for example: current work role, professional qualification and resources available to support staff when their role description includes research. Additional questions are also included for needs assessment purposes to explore barriers and enablers to research at both individual and team levels.	Individual, team and organisation levels.	Face validity ofthe RCC tool was assessed by an international expert to ensure all relevant aspects were included and measured appropriately. Face validity was further examined by a group of 12 university staff, both academic and administrative, who were asked to complete the survey and provide feedback. [...] + see methodology since validation was the main aim of the study.	Allied health professional	The RCC tool was administered to a sample of 134 allied health professionals. Approximately 1000 allied health workers were emailed and.	Not specified but before 2010	CB + definition also of research culture	Research capacity building (RCB) is defined as a process of developing sustainable abilities and skills enabling individuals and organisations to perform high quality research (Trostle 1992;D'Auria 2000).
Jule, A.; Furtado, T.; Boggs, L.; Loggerenberg, F. van; Ewing, V.; Vahedi, M.; Launois, P.; Lang, T.	2017	Developing a globally applicable evidence-informed competency framework to support capacity strengthening in clinical research.	BMJ Global Health	**Special Programme for Research and Training in Tropical Diseases (TDR) Global Competency Framework for Clinical Research.**	50 distinct competencies, which can be grouped into **five balanced categories**: Ethics, Quality and Risk Management; Study and Site(s) Management; Research Operations; Scientific Thinking; and Professional Skills.	Individual	In line with Lucia and Lepsinger's recommendations for the validation of competency frameworks,21 the pilot testing will follow	Applicable to all research roles	n/a	n/a	CD	Not given
Matus, J.; Wenke, R.; Hughes, I.; Mickan, S.	2019	Evaluation of the research capacity and culture of allied health professionals in a large regional public health service.	Journal of multidisciplinary healthcare	**Research Capacity and Culture (RCC) tool - Individual (15 questions)** Each question uses a 10-point numeric scale format where 1 is the lowest possible level of skill or success and 10 is the highest possible level of skill or success. There is also an option of "unsure" 15 minutes to complete.	The tool includes **52 questions that examine participants' selfreported success or skill in a range of areas related to research capacity or culture across three domains**, including the Organization (18 questions), Team (19 questions), and **Individual (15 questions).** The RCC tool also includes questions **about perceived barriers to and motivators for undertaking research.** Participants can indicate which, from a list of 18 barriers and 18 motivators, apply to themselves as individuals and within their team and to add any other items that are not included in this list. Finally, the tool includes questions about participants' demographics, work role, professional qualifications, and formal research engagement and what research activities are required and supported in their current work role.	Individual, team and organisation levels	Face validity ofthe RCC tool was assessed by an international expert to ensure all relevant aspects were included and measured appropriately. Face validity was further examined by a group of 12 university staff, both academic and administrative, who were asked to complete the survey and provide feedback. [...] + see methodology since validation was the main aim of the study.	All 875 AHPs and 97 allied health assistants employed by Gold Coast Health (GCH) were invited to participate in a survey consisting of the RCC tool. Students and non-AHPs including medical and nursing staff were excluded.	A total of 369 AHPs accessed the survey during the data collection period. Of them, four did not consent to participate in the survey, 63 provided consent but did not answer any of the questions, 44 incompletely answered the questions, and 258 completed the entire survey. In total, data from 302 AHPs were available for analysis, which represent 30% of the estimated total allied health workforce within our health service.	April' 2017	CB	Capacity building defined as a process of individual and institutional development which leads to higher levels of skills and greater ability to perform useful research (Trostle J. 1992).
Njelesani, J.; Dacombe, R.; Palmer, T.; Smith, H.; Koudou, B.; Bockarie, M.; Bates, I.	2014	A Systematic Approach to Capacity Strengthening of Laboratory Systems for Control of Neglected Tropical Diseases in Ghana, Kenya, Malawi and Sri Lanka.	PLoS Neglected Tropical Diseases	**Questionnaire for laboratory managers**, a **semi-structured interview guide** for use with laboratory stakeholders, a capacity gap checklist for use with the laboratory manager and laboratory staff, and a checklist for ISO 15189 to be used for on-site observations.	n/a	Individual, organisations levels.	We revised the tools after their implementation in each country by conducting a retrospective analysis ofhow the tools contributed or not to the awareness phase in the framework for designing and evaluating a health research capacity-building programme that guided the design of our capacity strengthening tools.	Laboratory scientists, laboratory directors, research staff, WHO staff, ministry representatives, students, human resource and financial staff, donors, and senior academics.	62 semi-structured interviews were conducted, 17 in Malawi, 11 in Ghana, 16 in Kenya, and 18 in Sri Lanka.	2012	CD	Capacity development defined as the ability of individuals, organisations or systems to perform appropriate functions effectively, efficiently and sustainably''(Milen A 2001)
Petersen, M.;Pain, T.; Fernando, M.	2018	Building Allied Health Research Capacity at a Regional Australian Hospital: A Follow-up Study.	Internet Journal of Allied Health Sciences & Practice	**Cross sectional electronic survey - 14 areas of expertese** responses were set out as **5-point Likert scales .**	The original survey was modified to capture additional information regarding motivators for participating in research. The data collected included **demographic** variables (age, gender, work-place characteristics), and **level of research experience and support needs**. Participants were also asked to rate the benefits of research and research barriers and enablers. All responses were set out as 5-point Likert scales.	Individual + motivation and barriers .	n/a	Allied health professionals	The response rate for the 2011 survey was 43% (n=248) while the 2015 survey achieved a 37% response rate (n=234 out of 618 AHP in the THHS)	The baseline survey was conducted in 2011 which was compared to a follow up survey in Nov 2014 - March 2015	CB	n/a
Raghavan, R.; Kupfer, L.	2009	Tracking trainees to success.	N/A	**Webbased trainee tracking tool – CareerTrac**	FIC grantees enter both **qualitative and quantitative information about their trainees including**: name, gender, country of origin, contact information, (...) training, accomplishments such as fellowships, awards (...), publications, funding received, and posters at scientific conferences, Career highlights, leadership, post-training competitive funding, presentations and consultations at scientific meeting. that are related to FIC training. This first of its kind system was developed as a tool for: 1-FIC staff to evaluate the effectiveness and impact of its health research training programs 2- FIC grantees to track their trainees and use the information in progress reports and when applying for other grants.	Individual	n/a	FIC trainees	About 3500 trainees in CareerTrac.	The CareerTrac application has been operational since December 2008.	n/a	n/a
Ried, K.; Farmer, E. A.; Weston, K. M.	2006	Setting directions for capacity building in primary health care: a survey of a research network	BMC Family Practice	**Research spider' tool - 10 areas of reseach with individual level of competency measure with a five point scale ranging from 1** (no experience or interest) **to 5** (high experience or interest).	Part three of the survey sought to ascertain the level of research experience using the visual 'research spider', a simple instrument for self-evaluation of knowledge. The 'research spider' was used to collect information on individual research experience and **interest in research skill development in ten core areas**. These include 'writing a research protocol', using quantitative research methods', 'publishing research', 'finding relevant literature', and 'applying for research funding' [all ten core areas are displayed in Figure 2]. In each area, the level of experience was measured on a five-point scale, from 1 (no experience) to 5 (high experience).	Individual	Smith H, Wright D, Morgan S, Dunleavey J: The 'Research Spider': a simple method of assessing research experience. Primary Health Care Research and Development 2002, 3:139-140.	PRIamry health care . The majority of respondents (55%) worked in allied health and general practice. 'Allied health' included consumer/community health, aged care, child, youth and women's health, chronic illness, mental health, nutrition, occupational therapy, pharmacy. General practice included divisional staff. Other professional areas were nursing, health services including coordinators, managers and information technologists, medical specialists, and academics.	A total of 89 members out ot 229 members (40% of all first year members) participated in the survey. More female health professionals (74%) than male health professionals joined SARNet in 2002/03, and accordingly represented the majority of respondents (76%). The median age of all respondents was 40, ranging from 21 to 62 years. .	The baseline survey was sent to all members who joined between November 2002 and December 2003.	CB	Capacity buildingdefined as a whole system approach, which promotes participation of new researchers through to more experienced practitioners.
Ried, K.; Farmer, E. A.; Weston, K. M.	2007	Bursaries, writing grants and fellowships: a strategy to develop research capacity in primary health care.	BMC Family Practice	**Semi-structured 40 minute telephone interview** conducted by an external and independent interviewee. **A copy of the research spider was available** to grant holders during the telephone interview. Grant holders were asked to rate their perceived skill level **from 1 = 'no experience' to 5** = 'very experienced' prior to and after their research activity. Grant recipients were **asked to rate the impact of the supported activity on their capacity and confidence to participate in or initiate a research project**, their confidence to seek collegial support for research collaboration, and their general interest in doing future research. Impact was **measured on a 5-point scale,** with a score of 1 reflecting 'no impact', and a score of 5 reflecting 'substantial impact'.	Feedback was sought on acquisition of research skills, publication outcomes, the impact on confidence, and interest in pursuing research in the future in relation to the PHCRED funded project. In total **10 areas** were searched. Figure 3: The 'research spider' was used to assess grant holders' research experience before and after the supported research activity. The level of experience was measured using a five point scale ranging from 1 (no experience) to 5 (very experienced).	Individual + motivation and barriers.	Smith H, Wright D, Morgan S, Dunleavey J: The 'Research Spider': a simple method of assessing research experience. Primary Health Care Research and Development 2002, 3:139-140. 11. Cooke J: A framework to evaluate research capacity building in health care. BMC Fam Pract 2005, 6:44.	Most of the grant holders were allied health professionals (47%). A smaller number were general practitioners (GPs) (21%), Division of General Practice staff (16%), and nursing professionals, medical and primary health care students, and one consumer organisation.	Thirty-four (89%) grant holders participated in the evaluation out of 38 PHCRED funding holders who had completed their projects by early 2006; 21 of 24 bursary holders (88%), 10 of 11 writing grant holders (91%), and 3 of 3 research fellows. The majority of Flinders PHCRED funding recipients lived in metropolitan Adelaide (73.5%), were female (74%), and between 35 and 54 years of age (79%).	2006	n/a	n/a
Sarre, G.; Cooke, J.	2009	Developing indicators for measuring Research Capacity Development in primary care organizations: a consensus approach using a nominal group technique.	Health & Social Care in the Community	**Indicators to measure RCD** in institutions. Starting from **Cooke 2005** - Nominal group technique use to improve and test the list.	**9 Principles:** The number of indicators agreed by consensus for eachprinciple ranged from 22 indicators for infrastructure to only four indicators each for those relating to research culture and continuity, and sustainability. Pricinples are: infrastructure, linkages and partnerships, dissemination, Skills development, Reseach activity, Close to practice, Leadership, Continuity and sustainability, Research Culture.	Institutional and Individual	No	Stage 1: NHS Trust R&D managers. Stage2: Purposive sample includes nine participants comprised three R&D managers, two research facilitators, two research practitioners (one allied health professional and one general practitioner), one RDSU member with mentoring and capacity-building expertise in primary care and one Primary Care Research Fellow.	Stage 1: over 70 participants who included NHS Trust R&D managers . Stage2: 9 out of 11 invited experts for this study, individuals with experience of research capacity building, doing research, mentoring researchers and those who had been on the receiving end of research capacity-building initiatives.	The study was undertaken between March 2005 and April 2006.	CD	Research Capcity development is 'a process of individual and institutional development which leads to higher levels of skills and greater ability to perform useful research' (Trostle 1992, p. 1321)
Sonstein, S. A.; Namenek Brouwer, R. J.; Gluck, W.; Kolb, H. R.; Aldinger, C.; Bierer, B. E.; Jones, C. T.	2018	Leveling the Joint Task Force Core Competencies for Clinical Research Professionals.	Therapeutic Innovation and Regulatory Science	**Joint Task Force for Clinical Trial Competency (JTF) Core Competecies Framework 2.0**. Use 3 levels of competencies. They are **(1) Fundamental** ("Can perform the task/and or exhibit the knowledge at an essential or fundamental level; may require some coaching or supervision"), **(2) Skilled** ("Can perform task or skill independently, consistently, accurately, and has a moderate level of expertise. Efficient and high-quality work; able to independently navigate resources and uses tools well"), and **(3) Advanced** ("Demonstrates advanced skills and knowledge and the ability to teach, coach, or supervise others. Consistently applies critical thinking and problem solving".	The JTF Framework is composed of **8 Domains (Figure 1) 1- Scientific Concepts and Research Design 2- Ethical and Participant Safety Considerations 3- Investigational Products Development and Regulation 4- Study Operations (Good Clinical Practice) 5- Study and Site Management 6 Data Management and Informatics 7-Leadership and Professionalism 8-Communications and Teamwork- and 47 core competency statements** that were aligned and harmonized utilizing published statements of core competency requirements to encompass the entire clinical research enterprise.	Individual	Ultimately, the profession of clinical research was recognized by the Commission on Accreditation of Allied Health Education Programs (CAAHEP). academic programs in clinical research are now eligible for CAAHEP accreditation based on JTF core competency-informed educational standards.	Applicable to all research roles	n/a	2016	CB	n/a
Swenson-Britt, E.; Berndt, A.	2013	Development and Psychometric Testing of the Nursing Research Self-Efficacy Scale (NURSES).	Journal of Nursing Measurement	**Nursing Research Self-Efficacy Scale (NURSES)** is a 38-item instrument using a **5-point Likert scale** format ranging from 1 (very little) to 5 (quite a lot). Time to complete the NURSES instrument would be about 15 min.	Measure individual nurses' degree of research self-efficacy and their perceptions regarding their unit's collective support of research use. The NURSES is a 38-**item Likert scale instrument** developed through ongoing instrument validation that included content validation and exploratory and confirmatory analysis. The instrument contained **5 subscales** include obtaining science-based knowledge resources, critically reading and evaluating quantitative research literature, critically reading and evaluating qualitative research literature, understanding and applying theory, and collective research efficacy.	Individual	An innovative conceptual framework, synthesizing organizational culture and leadership. This model is presented as a theoretical subscription using a modified Gibb's Framework (McQuiston & Campbell, 1997), theory (Schein, 2004) with social cognitive theory (Bandura, 1997), was the basis for this instrument development.	Nurses from multiple hospitals.	The total sample included 687 nurses who responded to the survey; of these nurses, 649 completed all sections out of the 2000 contacted.	In 4 phases between 2005 and 2008	Research Self-Efficacy	Swenson-Britt & Berndt, (2009) used the term research self-efficacy to describe the extent to which practicing nurses expressed confidence in their ability to understand and use research findings. Specifically, RSE measured the extent to which practicing nurses could (a) obtain research resources; (b) understand theory; (c) evaluate quantitative research; (d) evaluate qualitative research; and (e) apply research findings in their practice.
Torres, G. C. S.; Estrada, M. G; Sumile, E. F. R.; Macindo, J. R. B.;Maravilla, S. N.; Hendrix, C. C.	2017	Assessment of Research Capacity Among Nursing Faculty in a Clinical Intensive University in The Philippines.	Nursing Forum	**Modified University of Memphis (UM) questionnaire** contained **30 items** to measure the ability and motivation to do research among faculty members and reportedly took about 15 min to complete. **Of the roginal 41 items** some items were revised, and non relevant deleted. Lastly, the study team added new items to include self-assessment of knowledge and skill on the research process, including research designs, and items to assess perceived resources and barriers in conducting research at UST-CON. Perceived levels of research capacity were assessed using a **Likert scale, with 1 as the lowest possible score and 5 as the highest.**	**5 AREAS- 30 subareas:** subject demographics; 1- Conceptualization phase (5 subcategorie) 2- Design and planning phase (10 subcategories) 3- Empirical phase (6 Sub) - 4- Analytic phase (5 sub) 5 - Dissemination phase (4 sub)	Individual	Meyers, A., & Kurtz, H. (2013). University of Memphis research capacity assessment taskforce report. Retrieved October 21, 2014, from http://www.memphis.edu/researchsupport/pdfs/research capacity_assessment.pdf	Clinical and nonclinical faculty members who had a regular teaching assignment at UST-CON.	Of the 82 faculty at UST-CON, 66 participated (80.49%) in the study. Thirty faculty (45.45%) completed the survey electronically, while the rest chose the pen-and-paper method. faculty members.	The survey was conducted from August to October 2015.	Research capacity	Research capacity is the capability to conduct high-quality research undertakings and is crucial toward building the knowledge base for evidence based nursing practice (Sergott, McIvor, & Green, 2006).
Williams, C.; Miyazaki, K.; Borkowski, D.; McKinstry, C.; Cotchet, M.; Haines, T.	2015	Research capacity and culture of the Victorian public health allied health workforce is influenced by key research support staff and location	Australian Health Review	**The research capacity and culture (RCC) tool. 15 questions for Individal level (Table 4).** Each item used a 10-point numeric rating scale response format with 1 being the lowest and 10 being the highest success or skill leve	The tool contains **51 items examining participants' self-perceived skills or success in a range of areas related to research capacity or culture at the individual, team or organisational level.** Each item used a 10-point numeric rating scale response format with 1 being the lowest and 10 being the highest success or skill level.Aforced response format was used throughout the questionnaire to reduce missing data and the 'don't know' option was removed.	Individual, team and organisation levels	SELECTED PAPER ABOVE: Holden L, Pager S, Golenko X, Ware RS. Validation of the research capacity and culture (RCC) tool: measuring RCC at individual, team and organisation levels. Aust J Primary Health 2012; 18: 62–7. doi:10.1071/ PY10081	Allied health professions invited to participate were dietetics, drug and alcohol workers, exercise physiology, music therapy, occupational therapy, oral health (not dentistry), orthotics and prosthetics, physiotherapy, play therapy, podiatry,psychology, social work, speech pathology, audiology, orthoptics, pathology, pharmacy, radiation therapy, radiography and sonography.	Of the 539 allied health participantswhostarted the online survey, 520 fully completed the surveys.	The survey was open from 10 April to 1 June 2014.	n/a	n/a

Figure 2 - *PRISMA screening diagram*



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3/4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4/16
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6/7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7/8
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	8
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	10
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	11
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	9
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	n/a
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	14
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/a
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10/20/21
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	14
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	14
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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Measuring Research Capacity Development In Healthcare Workers: a systematic review.

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Measuring Research Capacity Development In Healthcare Workers: a systematic review.

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ABSTRACT

Title: Measuring Research Capacity Development In healthcare workers: a systematic review.

Objectives: A key barrier in supporting Health Research Capacity Development *(HRCD)* is the lack of empirical measurement of competencies to assess skills and identify gaps in research activities. An effective tool to measure Health Research Capacity Development in healthcare workers would help inform teams to undertake more locally-led research. The objective of this systematic review is to identify tools measuring healthcare workers' individual capacities to conduct research.

Design: Systematic review and narrative synthesis using PRISMA checklist for reporting systematic reviews and narrative synthesis and CASP checklist on critical appraisal for qualitative studies.

Data sources: 11 databases were searched from inception to 16 January 2020. The first 10 pages of Google Scholar results were also screened.

Eligibility Criteria: We included papers describing the use of tools/ to measure/assess Health Research Capacity Development at an individual level among healthcare workers involved in research. Qualitative, mixed and quantitative methods were all eligible. Search was limited to English language only.

Data extraction and synthesis: Two authors independently screened and reviewed studies using Covidence software, and performed quality assessments using the extraction log validated against the CASP qualitative checklist. The content method was used to define a narrative synthesis.

Results: The titles and abstracts for 7,474 unique records were screened and the full text of 178 references were reviewed. 16 papers were selected: seven quantitative studies; one qualitative study; and five mixed-method studies. Tools with different levels of accuracy in measuring Health Research Capacity Development in healthcare workers at the individual level were described. The Research Capacity and Culture Tool and the "Research Spider" tool were the most commonly defined. Other tools designed for ad hoc interventions with good generalizability potential were identified. Three papers described health research core competency frameworks. All tools measured Health Research Capacity Development in healthcare workers at an individual level with the majority adding a measurement at the team/organisational level, or data about perceived barriers and motivators for conducting health research.

Conclusions: Capacity building is commonly identified with pre-post intervention evaluations without using a specific tool. This shows the need for a clear distinction between measuring the outcomes of training activities in a team/organisation, and effective actions promoting Health Research Capacity Dvelopment. This review highlights the lack of globally applicable comprehensive tools to provide comparable, standardised and consistent measurements of research competencies.

Review registration and funding: PROSPERO (CRD42019122310). Independent research.

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ARTICLE SUMMARY

Strengths

- Thoroughly-conducted systematic review collecting data from all major existing databases and grey literature.
- Topic not previously addressed in other reviews searching for tools to measure health research capacity building at individual level.
- Brief overview of the identified tools to measure health research capacity building at individual level highlighting strengths and weaknesses of them.

Limitations

- Complex identification of relevant studies due to the lack of clarity on a common definition and terminology to identify Health Research Capacity Development (HRCD).
- None of the studies use the standard reporting procedures for qualitative or quantitative research.

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INTRODUCTION

In 2004 the Global Forum for Health Research highlighted the challenge for low and middle-income countries to have the capacity to perform effective and locally-led health research which address the major health problems affecting their own populations (1-3). Twenty years later, low and middle-income countries still carry 90% of the global disease burden, but only 10% of global funding for health research is devoted to addressing these persistent health challenges (4). Health research capacity development (HRCD) for healthcare workers has been recognised as a critical element to overcoming global health challenges, especially in low and middle-income countries (5). For too long HRCD in low and middle-income countries has been documented through training programmes which enable local teams to participate in externally sponsored trials, creating a false appearance of growth and generating dependence on foreign support (6, 7).

The process of progressive empowerment is usually referred to as capacity development (8). This term has been used in multiple areas and applied in different sectors to develop new or existing competencies, skills, strategies at a macro or individual level (9). In the field of health, research capacity development should support healthcare workers in generating local evidence-based results to inform policy and improve population health. The three health-related Millennium Development Goals, and more recently the targets "B" and "C" of the Sustainable Development Goals, all support the adoption of new strategies to strengthen the capacity of healthcare workers in all countries in performing their job and engaging in research (10-12). One of the critical barriers in supporting HRCD is the lack of empirical measurement of competencies in relation to the performance of research activities. Existing frameworks and tools have been developed for a particular purpose in a particular context (13) (14). Others have identified barriers healthcare workers encounter in engaging in research or have monitored and evaluated targeted training activities (15). This systematic review aims to identify tools to measure individual healthcare workers' capacities to conduct research.

METHODS

The PRISMA checklist (16) for reporting systematic reviews and narrative synthesis and CASP checklist (17) on critical appraisal for qualitative studies were used to design this systematic review and to refine the extraction log according to recognised guidelines.

Inclusion and exclusion criteria

The aim of the systematic review was to identify existing tools which measure individual capacities in conducting research in healthcare workers. The inclusion and exclusion criteria were defined in advance and documented using an adapted version of a SPIDER table (Table 1). The primary population of interest were all health-related professionals or healthcare workers involved in research activities. healthcare workers delivering health services when research was not considered as the focus of the study were excluded. Occupational health research was excluded. Studies about volunteers, defined as people offering their services to support health activities with no specific training as health professionals, were also excluded. Initially, only healthcare workers working in low and middle-income countries were included, but this limitation was removed to identify any tool measuring HRCD in any setting. The Phenomenon of Interest was defined as: assessing HRCD; or identifying tools, frameworks and templates designed to assess HRCD. A comprehensive range of terms including synonyms for "assess", "tool" or "development" was used. Studies were excluded which mentioned components that could be considered to assess, measure and 'give evidence to' research capacity development, but were not presented in any capacity development context. In addition, since the concept of capacity development is widely applied to different settings, studies on areas unrelated to health, such as "air pollution", "financial capacity" or "tobacco", were also excluded. The study design criteria were broad,

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to include qualitative, quantitative and mixed methods papers. Further criteria of eligibility included in the SPIDER table refers to the quality of the study (Evaluation) and the Research type.

Table 1 - ***SPIDER diagram - Inclusion and exclusion criteria***

SPIDER	*Inclusion*	Exclusion
Sample (population involved)	Healthcare workers involved in research defined as: Health professionals involved in research activities at every step of research and with any health-related professional profile working in health centres.	Healthcare workers who delivered health services when research is not considered People involved in occupational health research Volunteers defined as people offering their services to support health activities with no specific training as health professionals.
Phenomenon of Interest	Broadly; assessed health research capacity development. Specifically; analysed tools, frameworks, templates to assess and measure health research capacity development/building/strengthening. Key words: Tool – framework - assess – measure – give evidence - capacity development – capacity building – capacity strengthening - competency improvement – professional development - mentoring	Included specific components that could be considered as aiming to assess, measure and give evidence to research capacity development, but not presented in any capacity development context. Non-health-related capacity development research such as “air pollution”, “financial capacity”, “tobacco” or other areas unrelated to health research emerging from the search refining.
Design (nature of the intervention described)	Qualitative literature including interview, focus groups, surveys, grounded theory, ethnography, interpretative phenomenological analysis, content analysis, thematic analysis. Quantitative literature including score systems, competencies measurement. Mixed methods papers combining qualitative and quantitative approach	Studies that did not meet the inclusion criteria for design.
Evaluation	Acceptable: met high and moderate scores in quantitative assessment tool EPHPP and measured qualitative validity with COREQ evaluation method if applicable.	Studies that did not meet the inclusion criteria for evaluation.
Research Type	Qualitative, mix and quantitative methods	Did not have English language abstract

Information sources and search strategy

Eleven databases were searched from inception to 16 January 2020: Ovid Medline; Ovid Embase; Ovid PsycINFO; Ovid Global Health; Ebsco CINAHL; ProQuest Applied Social Sciences Index & Abstracts (ASSIA); ProQuest Sociological Abstracts; ProQuest Dissertations & Theses Global; Scopus; Web of Science Core Collection; and the WHO Global Index Medicus Regional Libraries. The first ten pages of results from Google Scholar were also screened. The search strategies used free-text terms and combinations of the relevant thesaurus terms, limited to English language publications only, to combine terms for capacity building, measuring, and health research. The“NOT” command was used to exclude papers about students, postgraduate students, tobacco, air pollution, and a variety of other concepts to minimise the number of irrelevant results (Box 1 for a full set of search strategies).

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BOX 1:

Database: Medline (Ovid MEDLINE® Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE® Daily and Ovid MEDLINE®) 1946 to present

Search Strategy:

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1 Capacity Building/ (1965)
2 (capacit* adj2 build*).ti,ab. (5789)
3 (capacit* adj2 develop*).ti,ab. (3591)
4 (capacit* adj2 strengthen*).ti,ab. (924)
5 (competenc* adj2 improv*).ti,ab. (1460)
6 ((professional* adj2 develop*) and (competenc* or capacit*)).ti,ab. (1747)
7 1 or 2 or 3 or 4 or 5 or 6 (13649)
8 Mentoring/ (820)
9 mentor*.ti,ab. (13369)
10 (assess* or measur* or evaluat* or analys* or tool* or equip*).ti,ab. (9653076)
11 "giv* evidence".ti,ab. (3814)
12 framework*.ti,ab. (231138)
13 8 or 9 or 10 or 11 or 12 (9763562)
14 Research/ (196782)
15 clinical.ti,ab. (3158817)
16 (health* and research*).ti,ab. (337604)
17 14 or 15 or 16 (3588891)
18 7 and 13 and 17 (3433)
19 18 (3433)
20 limit 19 to english language (3346)
21 (student* or graduate or graduates or postgraduate* or "post graduate*" or volunteer* or communit* or tobacco or "climate change" or "air pollution" or occupational or "financial capacity" or informatics or "IT system" or "information system" or transport or "cultural competenc*" or disabili* or trauma).ti,ab. (1828113)
22 20 not 21 (1673)

Google Scholar – Screen the first 10 pages of results

Sorted by relevance:

("capacit* build*"|"build* capacit*"|"capacit* develop*"|"develop* capacit*"|"capacit* strengthen*"|"strengthen* capacit*"|"professional* develop*"|"completenc* improv*"|"improv* competenc*")("health* research*"|clinical)

https://scholar.google.co.uk/scholar?q=(%22capacit*+build*%22%7C%22build*+capacit*%22%7C%22capacit*+develop*%22%7C%22develop*+capacit*%22%7C%22capacit*+strengthen*%22%7C%22strengthen*+capacit*%22%7C%22professional*+develop*%22%7C%22completenc*+improv*%22%7C%22improv*+competenc*%22)(%22health*+research*%22%7Cclinical)&hl=en&as_sdt=0,5

Study selection

Two researchers, DB and ER independently screened and reviewed studies using the Covidence systematic review software (18). In case of disagreement, DB and ER discussed the abstracts in question. After consensus on inclusion was reached, the full-texts of all included studies were rechecked for inclusion by DB and confirmed by ER.

Study analysis procedure

Data from selected papers were extracted, and quality assessments performed using an extraction log created and validated against the CASP checklist (17) on critical appraisal for qualitative studies. Macro areas of interest in the log were: general information on the paper such as author and title, main focus and study design. The source of funding, conflict of interests and ethics approval were also recorded. A separate section of the extraction log recorded the characteristics of the tool used or described in each selected paper (Figure 1). The extraction log also included specific sections considering the study design, the methodology and the main findings of each paper. Furthermore, a dedicated section of the

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log collected data on the quality of each study, analysing selection biases and a critical appraisal derived from the CASP checklist. If a definition of Capacity Development was given, the definition was collected. Some of these sections of the extraction log are not present in Figure 1 since it focuses on the description of the identified tool. The content method was used to define a narrative, described in the Discussion section.

RESULTS

Database search and results screening

In December 2018, the first round of the search was performed in 11 different databases and in Google Scholar using the search strategy described in box 1. A total of 13,264 suitable records were found. A total of 6,905 duplicates were removed, resulting in 6,359 unique records for inclusion screening by title and abstract (Table 2), which was performed throughout 2019. In January 2020 an additional search for papers published or included in publication databases in 2019 was performed using the same search strategy and resulted in 15,775 papers and after removal of duplications, a total of 1,118 papers were found. These papers were then added to the 6,359 papers identified from the first search. 7,474 unique papers were included for title and abstract inclusion screening (three duplicate records were removed in the Covidence software).

Table 2 - *Search Results*

	Search dates: 20-21/12/18	Search date: 16/01/2020
Ovid Medline (searched on 20/12/18)	1673	1937
Ovid Embase (searched on 20/12/18)	2344	2721
Ovid PsycINFO (searched on 20/12/18)	597	619
Ovid Global Health (searched on 20/12/18)	566	676
EBSCO CINAHL (searched on 20/12/18)	1376	1663
ProQuest Sociological Abstracts (searched on 20/12/18)	305	371
ProQuest ASSIA(searched on 20/12/18)	1463	2036
ProQuest Dissertations & Theses Global (searched on 20/12/18)	279	312
Scopus (searched on 21/12/18)	2230	2575
Web of Science – Core Collection (searched on 21/12/18)	1900	2195
WHO Global Index Medicus (searched on 21/12/18)	531	670
Total	13264	15775
Total after deduplication	6359	7469
16/01/2020: Unique to databases since 20-21/12/18		1118
TOTAL TO SCREEN		7477

The 7,474 unique relevant studies identified were uploaded to the Covidence systematic review software. Two researchers, DB and ER independently screened the studies, including or excluding according to the criteria in the SPIDER table (Table 1). A total 7280 studies were considered irrelevant. The full text papers for the remaining 178 references were reviewed. Reasons for exclusion were identified by streamlining the SPIDER table criteria into three main criteria: wrong setting, irrelevant study design and wrong focus of the study. A reason for exclusion was assigned to each paper. All 178 studies described some form of activity to measure the competencies related to performing health research. Thirty were excluded because they were literature reviews on a different aspect of health research or because they described a general perspective on the topic of health capacity development without offering any specific measurement or without reference to research. In addition, 42 studies were excluded because of the wrong setting, since competencies were measured at the level of research institutions or within a specific network. An additional 90

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studies were excluded because the study design did not match the inclusion criteria: 38 studies described the use of a measurement tool tailored to the context (e.g. specific profession, intervention or setting) and not at the individual level; the remaining 34 studies were excluded because there was no mention of a specific tool to measure HRCD. The final 18 papers reported the use of an evaluation tool, but the tool was an *ad hoc* pre-post intervention questionnaire with low potential of applicability in a context different from the one described in the paper. A total of 162 studies were therefore excluded, leaving 16 studies for this review (Figure 2).

Analysis of the findings across the selected papers

A total of 16 studies met the inclusion criteria set for this systematic review (19-34). The 16 articles were analysed using the extraction log created and validated against the CASP qualitative checklist.

The results are summarised in Table 2. None of the papers were published before 2006 and only 9 of them were published after 2014 (20, 21, 23-26, 31, 33, 34). The majority (n = 13) applied a tool in high-income settings (19, 20, 22-24, 26-32, 34). Seven papers described the use of tools in Australia (20, 22, 24, 26, 28, 29, 34), three in low and middle-income countries (one in Ghana, Kenya, Malawi, Sri Lanka (25), one in the Pacific Islands (21) and one in the Philippines (33)), one in Europe (Norway) (19), one in the United States (32), and one measured HRCD in a group linked to a specific intervention located in multiple areas of the world. Three of them described the creation of a tool without applying it to any specific context (23, 30, 31), but they were all designed by research groups in high-income countries (one in the United States and two in the United Kingdom).

All of the selected studies applied quantitative, qualitative and mixed-method analyses. The preferred approach (n=7) was to generate quantitative data using an HRCD tool (20, 24, 26-28, 33, 34). One-third of the studies (n = 5) used a mixed methods approach (19, 21, 22, 25, 29); quantitative tools were associated with semi-structured interviews, or in some cases qualitative questions were added to the questionnaire. The three studies describing the creation of a tool were not analysed under this methodological category.

Of the 16 selected studies, three used the term “capacity development” (23, 25, 30), and two included a definition of the concept (25, 30). Seven papers used “capacity building” (20-22, 24, 26, 28, 31), of which four also included a definition (20, 22, 24, 28). In two papers, the capacity building definition was associated with the definition of “research culture” (20, 22). Two additional papers used alternative generic terms like “research capacity” (33) or “research self-efficacy” (32). Four papers did not refer to any specific term and therefore no definition was given (19, 27, 29, 34).

Five of the 16 selected papers openly declared no conflict of interests (22-24, 28, 31). Eight stated the source of funding used to carry out the activities described (19, 21, 23, 27-31). The number of participants in the studies varied from 28 enrolled participants for a qualitative study (21) to 3500 users of an online measurement tool (27).

Analysis of the tools from the selected papers

The tools described or used in the 16 selected papers varied in nature, length and applicability. In general, even when there were similarities, each paper described a different perspective on the use of a tool. Four papers applied a questionnaire-type tool to assess research competencies and skills (19, 21, 25, 33). The length of these questionnaires varied from 19 (21) to 59 (19) health research capacity

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related questions, with the addition of open-ended qualitative questions in two studies (19, 21), and a structured interview in another study (25).

Three studies (22, 24, 34) used, with a range of adaptations, the Research Capacity and Culture Tool and one study (20) revised this tool into a Research Capacity and Context Tool referencing Research Capacity and Culture tool as a primary source. Another recurrence in the papers was the use of the "Research Spider" tool (28, 29). Again, the original tool had been adapted to the context, and in one case (29), the tool was used as a base for qualitative research on HRCD. Two additional papers described tools designed *ad hoc* to measure the impact of an intervention (CareerTrac (27) and Cross-Sectional Electronic Survey (26)). These last two papers were not excluded under pre-post intervention since the action was wider, at a programme level, and the tool used to measure HRCD was the main focus of the paper. Furthermore, another paper described a tool for a specific category of healthcare workers (Nursing Research Self-Efficacy Scale -NURSES) (32). Three papers (23, 30, 31) focussed on the creation of a new tool and described the process of identifying a set of competencies required to run health research. The outcome of two of them was defined as a "core competency framework" (23, 31). The third defined the outcome of the analysis as a "set of indicators" (30).

In terms of the target population, the identified tools aimed to measure HRCD in a range of different healthcare worker professions. One-third of the papers (n=5) focused on measuring HRCD on Allied Health Professionals (AHPs) (20, 22, 24, 26, 34). Nurses were the main focus in two other studies (19, 32), and four studies applied a tool to a range of health professions (ranging from laboratory scientists to data managers) (21, 25, 28, 29). Two other papers focused on groups linked to a specific intervention (27, 33). All 16 papers included, alongside healthcare workers, representatives of technical professions in health such as managers, directors, faculty members, consumer organisation representatives. In the case of the three papers describing the creation of a new tool, they suggest that these tools would be applicable to all research roles (23, 30, 31).

As per inclusion criteria, the main level of measurement of the tools was at the individual level. Seven papers only measured HRCD at the individual level (19, 23, 28, 29, 31-33). Three papers added to the individual level of measurement by including information on the perceived barriers in performing health research (21, 26, 29); of these three, two also focused on understanding what motivates healthcare workers to become involved in health research (26, 29). The five studies, which used the Research Capacity and Culture tool and its variants, included the measurement of HRCD not only at the individual level, but also at the team and organisational level (20, 22, 24, 25, 34). One paper described the creation of a tool designed to be used at the organisational level, but embedded a measurement of HRCD at the individual level as well (30).

The most common way a selected tool was validated was by referencing the main paper that described the selected tool and its validation process (n = 6) (23, 28, 29, 32-34). This was the case for some of the *ad hoc* questionnaires (23, 33, 34), of the "Research Spider" tool (28, 29), and of the NURSES tool (32). Papers which described an original process or used modified versions of an original tool validated the tool through a contextual validation process described in the paper (21, 22, 24, 25, 31). These validation processes included a consultation of a panel of experts (22, 24, 31) or a reiterative process of validity (21, 25). One paper stated that the tool used was a validated tool without referencing the process or tool (20).

Overall, only two papers (23, 31) focussed specifically on tools to measure HRCD on a wider level, without linking the measurement to a specific group or a geographical area which was done in the

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majority of papers (19, 24, 25, 28, 29, 33). In four cases the tools described were adapted to identify determinants or barriers of HRCD in a defined setting (20, 30, 34) or to promote HRCD in relation to a specific disease or research topic (21). In other cases, the papers focussed on a tool aiming to assess the impact of specific interventions or programmes on HRCD (26, 27).

DISCUSSION

Summary of evidence

This systematic review aimed to identify tools which measure individual capacities in conducting research in healthcare workers; the 16 included articles (19-34) which demonstrated that tools to measure HRCD in healthcare workers are available, even if they are limited in number. In most cases, the identified tools do not originate from the need to measure and foster HRCD as a necessary strategy to promote research capacity. There is, therefore, a need to design more comprehensive tools which are globally applicable and able to provide comparable, standardised and consistent measurements of research competencies.

The importance of measuring HRCD has only been recognised recently (15). From the date of publication of the identified papers show, the contribution that health research can offer in capacity development at a personal level only begun in the first decade of this new millennium. Almost half of the selected papers (n= 7) refer to studies whose data have been collected after 2014 (20, 21, 24, 26, 31, 33, 34). Of note is the high number of new publications which were retrieved from the academic databases (1,118 papers) when the search strategy was re-run in 2020.

Questionnaires were the most commonly used method for assessing research skills and competencies. Almost two-thirds of the papers (n = 10) (19, 20, 22, 24, 26, 28, 29, 32-34) based the measuring system of different research skills at a personal level using a 5 point Likert scale (n = 6) (19, 26, 28, 29, 32, 33) or a 10 point scale (n = 4) (20, 22, 24, 34). This choice highlights the need for a validated quantitative tool based on a set of competency related questions that can bring standardisation, comparability and consistency across different roles and contexts. However, the extensive use of mixed methods, combining quantitative questionnaires with other qualitative instruments, reflects that HRCD depends on a complex series of components that need to be identified both qualitatively and quantitatively.

By not limiting the selection of articles for this review to those tools used in low and middle-income countries, this review has revealed that most of the tools identified were used in high-income settings. It is important to note that excluding pre/post-intervention assessments significantly reduced the inclusion of studies performed in low and middle-income countries. This finding highlights that although health systems in low and middle-income countries may benefit from providing evidence for HRCD (5), they are rarely the focus of the HRCD literature. Most of the measurements of HRCD in lower-income settings appears, in fact, to be narrowly linked to the measurement of the effectiveness of training offered for a specific study or limited to a particular disease. Even when the perspective is broader than a particular study, it is mostly limited to the evaluation and sustainability of training programmes and not linked to a plan of career progression and research competency acquisition. More attention should therefore be given in creating tools which are able to measure, support and promote long-lasting research capabilities in the perspective of professional growth for healthcare workers.

Three essential findings of this systematic review support a change in the perception of HRCD and the tools needed to measure it. Firstly, many of the excluded papers (42 out of 162 excluded papers from the last round of analysis) focused exclusively at the institutional level of measuring research capacity.

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This is mostly because training interventions are designed to prepare a team to run a study and rarely to promote individual HRCD (1, 35, 36). In some cases, the measurement via a tool is also an exercise to demonstrate the investment in training activities for reporting purposes (37, 38). It is therefore important to start promoting a more effective research culture which is independent from specific diseases or roles. This progression could be achieved by championing systems which measure the changes in research capacities at a team and personal level using a globally applicable tool. Most of the tools excluded were evaluation tools designed for, or used in a specific setting and thus not suitable for a comparable, standardised and consistent analyses of long term research competency acquisition strategies.

Secondly, papers that focused on measuring HRCD at the individual level confirmed that research is seen as an opportunity to learn the cross-cutting skills needed in healthcare. A defined set of standardised competencies required to conduct research could be used to measure an individual, team and organisation's abilities. This was the focus of two papers (23, 31) which identified a framework of core competencies. Most of the tools (n = 7) were designed to be applied to a wider variety of health professions (21, 23, 25, 28-31). HRCD can be accessed at different entry points depending on the specific job title, but the set of skills acquired is common and shared among the research team (1). The approach on assessing these inter-related competencies should therefore be global and not role or disease-based (39). The measurement at an individual level is essential to promote a consistent and coherent career progression for each person and role (40). However, the overall capability in running research programmes should be measured at a team level where all roles and competencies complement each other, skills are made visible, and measurable as a whole against an overall competency framework. Individual and institutional/team levels are therefore, two aspects of HRCD that grow together supported by a common comparable, standardised and consistent tool.

Thirdly, the lack of a standard definition for HRCD can lead to post-training evaluations being categorised as HRCD activities. Although pre/post training evaluations are important, it might be helpful to define what a 'structured action' is to promote HRCD. As previously mentioned, the term "capacity development" is not universally used, with many synonyms such as "research capacity" or "capacity strengthening", creating the possibility of different interpretations. Furthermore, inconsistent terminology was found in describing activities in support of HRCD that in reality, were very similar (e.g. workshop, training, course). Steiner *et al.* (41) suggest that there should be a standard definition in the context of educational capacity development. This suggestion, alongside a common taxonomy to describe health professions, would support the identification of HRCD as a defined process with specific characteristics and not with a general effort for research training.

The most common tool identified in this review was the Research Capacity and Culture Tool (20, 22, 24, 34). The Research Capacity and Culture tool consists of 52 questions that examine participants' self-reported success or skill in a range of areas related to research capacity or culture across three domains including the organization (18 questions), team (19 questions), and individual (15 questions). The Research Capacity and Culture tool includes questions on perceived barriers and motivators for undertaking research. The respondents of the Research Capacity and Culture are asked to rate a series of statements relevant to these three domains on a scale of 1–10, with one being lowest and ten being the highest possible skill or success level. It represents a good example of a comprehensive tool. As confirmed by the review findings, a potential limitation is its application mainly in an Australian context and almost exclusively to measure HRCD in AHPs (22, 24, 34). The generalizability of the tool should thus be confirmed. Nevertheless, the Research Capacity and Culture represents a strong example of

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how having a tool refined around a context, and a specific health profession can be an incentive in measuring HRCD.

Another tool highlighted by this review was the "Research Spider" tool (28, 29, 42). This tool collects information on individual research experience and interest in research skill development in ten core areas. These include "writing a research protocol", "using quantitative research methods", "publishing research", "finding relevant literature", and "applying for research funding". In each area, the level of experience is measured on a five-point Likert scale, from 1 (no experience) to 5 (high experience). The primary aim of the "Research Spider" is to be a flexible tool. This flexibility is confirmed in two studies (28, 29) which used the "Research Spider", with one (28) using it as the main measurement, and the other (29) as a quantitative base for qualitative semi-structured interviews. The advantage of this tool is that it provides a visual overview of personal research competencies. However, although the limited number of measurement areas (n = 10) makes the tool a good initial evaluation instrument, it does not offer a specification of the sub-skills of each area.

A critical mention should be reserved for the two papers which described the creation of a comprehensive research core competency framework (23, 31). Despite no specific tool being described and the competency scores being visualised by using a spider diagram, these studies present the most accurate overview of the skills required in running research programmes related to health. As mentioned before, a tool which applies a scoring system to the list of competencies identified by these frameworks has the potential of being widely applicable and reliable. This wide applicability and the absence of explicit biases in measuring research skills improvement, can foster a more robust approach to research in health. The measurement of HRCD unrelated to specific interventions would maximise the benefit of research at every level. At a personal level, it would clarify a potential career progression path highlighting possible gaps; at the team level, it would support a multidisciplinary approach to health challenges; and at an institutional level, the measurement of HRCD would make the know-how generated by the international scientific community accessible to a broader group of local health workers. Overall, health practice at a global scale would benefit from the incentive of getting involved in research derived from measuring the impact of it on improving competencies. Thus positive outcomes of measuring HRCD could place the issue of universal transferability, and applicability of research methodology and results at a higher level of priority in the design of health research projects.

Limitations of the Systematic Review

Methodological limitations are recognised for this systematic review. Firstly, there is a lack of clarity on a common definition and terminology to identify HRCD which complicates the search strategy. A long reiteration process was necessary when developing the search strategies for the databases to try and include all the possible variants used to define "tool", "capacities" and "development". Despite this effort, some studies may have been missed. Secondly, there was a lack of studies which referenced a standard reporting procedure, despite the presence of standards available for reporting qualitative or quantitative research (43-45) as well as for mixed-method research (46). Other limitations typical for reviews may also apply. Thirdly while this review has attempted to be as comprehensive as possible, some sources might not have been detected due to the challenge in finding all the relevant grey literature, and the restriction to English language sources only. Finally, it was not possible to analyse the psychometric aspects of each identified tool due to inconsistent reporting.

CONCLUSIONS

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Sixteen studies using or describing tools to measure HRCD were identified and analysed in this systematic review (19-34). Identifying capacity development with pre-post intervention evaluations or to generically evaluate capacity development without using a tool was common. ~~Furthermore, the majority of papers were not included due to measuring HRCD at an institutional level and not at an individual level~~. There is a need for a clear distinction between simply measuring training activity outcomes in healthcare workers and effective action promoting HRCD for healthcare workers.

The most recurrent tools described were the Research Capacity and Culture Tool (20, 22, 24, 34) and . the "Research Spider" tool (28, 29). A variety of other tools, mostly questionnaire-based, were identified, and in most cases, a broader applicability than described in the specific context of the paper may be possible. Two frameworks systematising research core competencies were identified (23, 31). The potential of tools derived from these frameworks could be significant. The applicability of each tool depends on the context and on the level of accuracy needed. Such tools could be routinely incorporated into standard personal development reviews in order to consistently support capacity development in research studies and organisations.

Future directions for HRCD include the design of a standardised, comparable and consistent tool to measure individual HRCD not linked to training evaluation, but support a long term research competencies acquisition strategy. In addition, the harmonisation of definitions and terminologies used in identifying HRCD actions and processes could facilitate standardisation and comparability of HRCD strategies.

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Data sharing:
All data relevant to the study are included in the article. Data are available upon reasonable request.

Competing interests
The authors declare that they have no competing interests.

Funding
No dedicated funds from any funder were allocated to support the study design; in the collection, analysis and interpretation of the data; in the writing of the report; and in the decision to submit the paper for publication.

Review registration
The protocol for this systematic review has been registered with PROSPERO. The registration number is CRD42019122310.

Ethics
No ethical approval was needed to conduct the study

Patient and Public Involvement
Due to the nature of systematic reviews, it was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

Authors' contributions
DB and ER designed and conducted the systematic review. DB wrote the draft of the systematic review and revised it according to the commentaries of ER, SB and TL. DB provided the final version of the manuscript. ER critically reviewed the manuscript and substantially contributed to the final version of the manuscript. SB critically reviewed both the design of the systematic review as well as the manuscript, and was involved in the development of meaningful inclusion criteria. TL critically reviewed the design of the study, made important suggestions for improvement, critically reviewed the manuscript and substantially contributed to the final version of the manuscript. All authors approved the final version of the manuscript.

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Figure1: Extraction log

Figure 2 - PRISMA screening diagram

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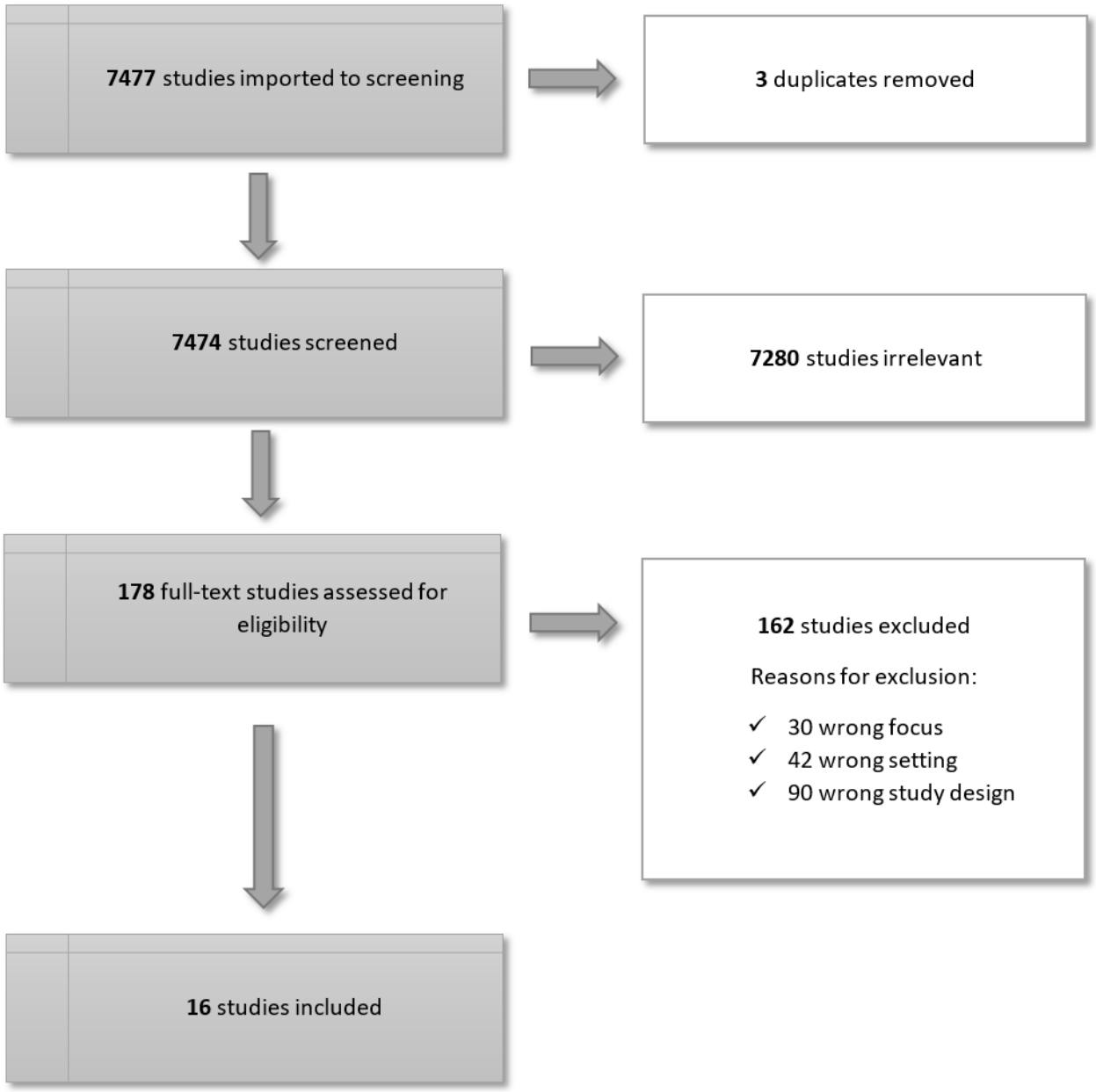
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Figure 1 - Extraction log

General Informations				Tool specification							Definitions	
Author s	Year	Title	Journal	Type of tool		Level of measurment	Validation of the tool	Study population	Number of participants (registered and completed)	Year of data collection	Variant defined (CB, CD, CS)	Definition given
Akerjordet, K.; Lode, K.; Severinsson, E.	2012	Clinical nurses' research capacity in a Norwegian university hospital: part 2.	Journal of Nursing Management	**Questionnaire from Australian experience** (Gethinget al 2001) -**59 questions** rated with five point Likert scale + open-ended questions .		Individual	No	nurses	364 out of 600 partecipants (61% response rate).	2011	n/a	Not given
Alison, J. A.;Zafiropoulos, B.;Heard, R.	2017	Key factors influencing allied health research capacity in a large Australian metropolitan health district.	Journal of multidisciplinary healthcare	**Research Capacity in Context Tool (RCCT)** - validated survey .Three domains - contains 18 questions for the organization domain, 19 questions for team domain, and **14 questions for the individual domain**. Rate on a 1–10-point Likert-style scale if the organization, the team, or they as individuals have success or skills in the features described in each item (where 1=no success/skill and 10=high success/skill).	RCCT survey measures research capacity and culture across three domains: organization, team, and individual with the **following AREAS**: 1) skills or success in research at the organization, team (or department), and individual levels; 2) barriers and motivators to performing research; 3) individual research activities completed in the past 12 months; 4) basic information regarding the individual's work role, qualifications obtained (ie, certificate, undergraduate, postgraduate, and PhD), and enrollment in higher degree study (yes/no) or other professional development related to research.	Individual, team and organisation levels.	Yes	Allied health professionals working within the Sydney Local Health District (SLHD	278 out of 514 contacte returned surveys giving a response rate of 54%.	Between November 2015 and May 2016	CB + definition also of research culture and research capacity	Research capacity building describes a process of individual or institutional development which leads to a higher level of skill, with a greater ability to perform research that promotes the health of individuals and communities.
Ekeroma, A. J.; Kenealy, T.; Shulruf, B.; McCowan, L. M.; Hill, A.	2014	Building reproductive health research and audit capacity and activity in the Pacific Islands (BRRACAP) study: methods, rationale and baseline results.	BMC Medical Education	**'Characteristics Questionnaire'** . The questionnaire was developed using a slightly modified tool that was validated in in Australia and made it applicable for the Pacific clinicans. **19 score questions + several opened questions.**	The questionnaire sought demographic professional and personal data that will inform of the participants' **research experience**, their current qualifications, work role and research expectations, and what they perceived to be **barriers** to and enablers of research in their setting. Specific question on **audit skills** are included. The questionnaire was piloted and could be completed comfortably within 30 minutes with responses collated thematically.	Individual + barriers.	The tool was tested using cognitive interviewing of five clinicians	Reproductive Health Clinicians	The 28 participants included 9 nurses/midwives, 17 medical doctors of whom 8 were specialists in reproductive health and 2 other health workers.	From feb 2013 to agust 2014	CB	Not given
Holden, L. ; Pager, S.; Golenko, X.; Ware, R. S.	2012	Validation of the research capacity and culture (RCC) tool: measuring RCC at individual, team and organisation levels.	Australian Journal of Primary Health	**The research capacity and culture (RCC) tool. 15 questions for Individal level (Table 7). Respondents rate on a scale of 1–10.**	Measure indicators of research capacity at individual, team and organisation levels. It **contains a series of statements relevant** to these three domains that **respondents rate on a scale of 1–10**, with one being lowest skill or success level and 10 being highest possible skill or success level. Items are scored separately for each domain. In addition to the scored items described above, **additional questions provide contextual information** to assist in the interpretation of the outcome scores, for example: current work role, professional qualification and resources available to support staff when their role description includes research. Additional questions are also included for needs assessment purposes to explore barriers and enablers to research at both individual and team levels.	Individual, team and organisation levels.	Face validity ofthe RCC tool was assessed by an international expert to ensure all relevant aspects were included and measured appropriately. Face validity was further examined by a group of 12 university staff, both academic and administrative, who were asked to complete the survey and provide feedback. [...] + see methodology since validation was the main aim of the study.	Allied health professional	The RCC tool was administered to a sample of 134 allied health professionals. Approximately 1000 allied health workers were emailed and.	Not specified but before 2010	CB + definition also of research culture	Research capacity building (RCB) is defined as a process of developing sustainable abilities and skills enabling individuals and organisations to perform high quality research (Trostle 1992;D'Auria 2000).
Jule, A.; Furtado, T.; Boggs, L.; Loggerenberg, F. van; Ewing, V.; Vahedi, M.; Launois, P.; Lang, T.	2017	Developing a globally applicable evidence-informed competency framework to support capacity strengthening in clinical research.	BMJ Global Health	**Special Programme for Research and Training in Tropical Diseases (TDR) Global Competency Framework for Clinical Research.**	50 distinct competencies, which can be grouped into **five balanced categories**: Ethics, Quality and Risk Management; Study and Site(s) Management; Research Operations; Scientific Thinking; and Professional Skills.	Individual	In line with Lucia and Lepsinger's recommendations for the validation of competency frameworks,21 the pilot testing will follow	Applicable to all research roles	n/a	n/a	CD	Not given
Matus, J.; Wenke, R.; Hughes, I.; Mickan, S.	2019	Evaluation of the research capacity and culture of allied health professionals in a large regional public health service.	Journal of multidisciplinary healthcare	**Research Capacity and Culture (RCC) tool - Individual (15 questions)** Each question uses a 10-point numeric scale format where 1 is the lowest possible level of skill or success and 10 is the highest possible level of skill or success. There is also an option of "unsure" 15 minutes to complete.	The tool includes **52 questions that examine participants' selfreported success or skill in a range of areas related to research capacity or culture across three domains**, including the Organization (18 questions), Team (19 questions), and **Individual (15 questions).** The RCC tool also includes questions **about perceived barriers to and motivators for undertaking research.** Participants can indicate which, from a list of 18 barriers and 18 motivators, apply to themselves as individuals and within their team and to add any other items that are not included in this list. Finally, the tool includes questions about participants' demographics, work role, professional qualifications, and formal research engagement and what research activities are required and supported in their current work role.	Individual, team and organisation levels	Face validity ofthe RCC tool was assessed by an international expert to ensure all relevant aspects were included and measured appropriately. Face validity was further examined by a group of 12 university staff, both academic and administrative, who were asked to complete the survey and provide feedback. [...] + see methodology since validation was the main aim of the study.	All 875 AHPs and 97 allied health assistants employed by Gold Coast Health (GCH) were invited to participate in a survey consisting of the RCC tool. Students and non-AHPs including medical and nursing staff were excluded.	A total of 369 AHPs accessed the survey during the data collection period. Of them, four did not consent to participate in the survey, 63 provided consent but did not answer any of the questions, 44 incompletely answered the questions, and 258 completed the entire survey. In total, data from 302 AHPs were available for analysis, which represent 30% of the estimated total allied health workforce within our health service.	April' 2017	CB	Capacity building defined as a process of individual and institutional development which leads to higher levels of skills and greater ability to perform useful research (Trostle J. 1992).
Njelesani, J.; Dacombe, R.; Palmer, T.; Smith, H.; Koudou, B.; Bockarie, M.; Bates, I.	2014	A Systematic Approach to Capacity Strengthening of Laboratory Systems for Control of Neglected Tropical Diseases in Ghana, Kenya, Malawi and Sri Lanka.	PLoS Neglected Tropical Diseases	**Questionnaire for laboratory managers**, a **semi-structured interview guide** for use with laboratory stakeholders, a capacity gap checklist for use with the laboratory manager and laboratory staff, and a checklist for ISO 15189 to be used for on-site observations.	n/a	Individual, organisations levels.	We revised the tools after their implementation in each country by conducting a retrospective analysis ofhow the tools contributed or not to the awareness phase in the framework for designing and evaluating a health research capacity-building programme that guided the design of our capacity strengthening tools.	Laboratory scientists, laboratory directors, research staff, WHO staff, ministry representatives, students, human resource and financial staff, donors, and senior academics.	62 semi-structured interviews were conducted, 17 in Malawi, 11 in Ghana, 16 in Kenya, and 18 in Sri Lanka.	2012	CD	Capacity development defined as the ability of individuals, organisations or systems to perform appropriate functions effectively, efficiently and sustainably''(Milen A 2001)
Petersen, M.;Pain, T.; Fernando, M.	2018	Building Allied Health Research Capacity at a Regional Australian Hospital: A Follow-up Study.	Internet Journal of Allied Health Sciences & Practice	**Cross sectional electronic survey - 14 areas of expertese** responses were set out as **5-point Likert scales .**	The original survey was modified to capture additional information regarding motivators for participating in research. The data collected included **demographic** variables (age, gender, work-place characteristics), and **level of research experience and support needs**. Participants were also asked to rate the benefits of research and research barriers and enablers. All responses were set out as 5-point Likert scales.	Individual + motivation and barriers .	n/a	Allied health professionals	The response rate for the 2011 survey was 43% (n=248) while the 2015 survey achieved a 37% response rate (n=234 out of 618 AHP in the THHS)	The baseline survey was conducted in 2011 which was compared to a follow up survey in Nov 2014 - March 2015	CB	n/a
Raghavan, R.; Kupfer, L.	2009	Tracking trainees to success.	N/A	**Webbased trainee tracking tool – CareerTrac**	FIC grantees enter both **qualitative and quantitative information about their trainees including**: name, gender, country of origin, contact information, (...) training, accomplishments such as fellowships, awards (...), publications, funding received, and posters at scientific conferences, Career highlights, leadership, post-training competitive funding, presentations and consultations at scientific meeting. that are related to FIC training. This first of its kind system was developed as a tool for: 1-FIC staff to evaluate the effectiveness and impact of its health research training programs 2- FIC grantees to track their trainees and use the information in progress reports and when applying for other grants.	Individual	n/a	FIC trainees	About 3500 trainees in CareerTrac.	The CareerTrac application has been operational since December 2008.	n/a	n/a
Ried, K.; Farmer, E. A.; Weston, K. M.	2006	Setting directions for capacity building in primary health care: a survey of a research network	BMC Family Practice	**Research spider' tool - 10 areas of reseach with individual level of competency measure with a five point scale ranging from 1** (no experience or interest) **to 5** (high experience or interest).	Part three of the survey sought to ascertain the level of research experience using the visual 'research spider', a simple instrument for self-evaluation of knowledge. The 'research spider' was used to collect information on individual research experience and **interest in research skill development in ten core areas**. These include 'writing a research protocol', using quantitative research methods', 'publishing research', 'finding relevant literature', and 'applying for research funding' [all ten core areas are displayed in Figure 2]. In each area, the level of experience was measured on a five-point scale, from 1 (no experience) to 5 (high experience).	Individual	Smith H, Wright D, Morgan S, Dunleavey J: The 'Research Spider': a simple method of assessing research experience. Primary Health Care Research and Development 2002, 3:139-140.	PRIamry health care . The majority of respondents (55%) worked in allied health and general practice. 'Allied health' included consumer/community health, aged care, child, youth and women's health, chronic illness, mental health, nutrition, occupational therapy, pharmacy. General practice included divisional staff. Other professional areas were nursing, health services including coordinators, managers and information technologists, medical specialists, and academics.	A total of 89 members out ot 229 members (40% of all first year members) participated in the survey. More female health professionals (74%) than male health professionals joined SARNet in 2002/03, and accordingly represented the majority of respondents (76%). The median age of all respondents was 40, ranging from 21 to 62 years. .	The baseline survey was sent to all members who joined between November 2002 and December 2003.	CB	Capacity buildingdefined as a whole system approach, which promotes participation of new researchers through to more experienced practitioners.
Ried, K.; Farmer, E. A.; Weston, K. M.	2007	Bursaries, writing grants and fellowships: a strategy to develop research capacity in primary health care.	BMC Family Practice	**Semi-structured 40 minute telephone interview** conducted by an external and independent interviewee. **A copy of the research spider was available** to grant holders during the telephone interview. Grant holders were asked to rate their perceived skill level **from 1 = 'no experience' to 5** = 'very experienced' prior to and after their research activity. Grant recipients were **asked to rate the impact of the supported activity on their capacity and confidence to participate in or initiate a research project**, their confidence to seek collegial support for research collaboration, and their general interest in doing future research. Impact was **measured on a 5-point scale,** with a score of 1 reflecting 'no impact', and a score of 5 reflecting 'substantial impact'.	Feedback was sought on acquisition of research skills, publication outcomes, the impact on confidence, and interest in pursuing research in the future in relation to the PHCRED funded project. In total **10 areas** were searched. Figure 3: The 'research spider' was used to assess grant holders' research experience before and after the supported research activity. The level of experience was measured using a five point scale ranging from 1 (no experience) to 5 (very experienced).	Individual + motivation and barriers.	Smith H, Wright D, Morgan S, Dunleavey J: The 'Research Spider': a simple method of assessing research experience. Primary Health Care Research and Development 2002, 3:139-140. 11. Cooke J: A framework to evaluate research capacity building in health care. BMC Fam Pract 2005, 6:44.	Most of the grant holders were allied health professionals (47%). A smaller number were general practitioners (GPs) (21%), Division of General Practice staff (16%), and nursing professionals, medical and primary health care students, and one consumer organisation.	Thirty-four (89%) grant holders participated in the evaluation out of 38 PHCRED funding holders who had completed their projects by early 2006; 21 of 24 bursary holders (88%), 10 of 11 writing grant holders (91%), and 3 of 3 research fellows. The majority of Flinders PHCRED funding recipients lived in metropolitan Adelaide (73.5%), were female (74%), and between 35 and 54 years of age (79%).	2006	n/a	n/a
Sarre, G.; Cooke, J.	2009	Developing indicators for measuring Research Capacity Development in primary care organizations: a consensus approach using a nominal group technique.	Health & Social Care in the Community	**Indicators to measure RCD** in institutions. Starting from **Cooke 2005** - Nominal group technique use to improve and test the list.	**9 Principles:** The number of indicators agreed by consensus for eachprinciple ranged from 22 indicators for infrastructure to only four indicators each for those relating to research culture and continuity, and sustainability. Pricinples are: infrastructure, linkages and partnerships, dissemination, Skills development, Reseach activity, Close to practice, Leadership, Continuity and sustainability, Research Culture.	Institutional and Individual	No	Stage 1: NHS Trust R&D managers. Stage2: Purposive sample includes nine participants comprised three R&D managers, two research facilitators, two research practitioners (one allied health professional and one general practitioner), one RDSU member with mentoring and capacity-building expertise in primary care and one Primary Care Research Fellow.	Stage 1: over 70 participants who included NHS Trust R&D managers . Stage2: 9 out of 11 invited experts for this study, individuals with experience of research capacity building, doing research, mentoring researchers and those who had been on the receiving end of research capacity-building initiatives.	The study was undertaken between March 2005 and April 2006.	CD	Research Capcity development is 'a process of individual and institutional development which leads to higher levels of skills and greater ability to perform useful research' (Trostle 1992, p. 1321)
Sonstein, S. A.; Namenek Brouwer, R. J.; Gluck, W.; Kolb, H. R.; Aldinger, C.; Bierer, B. E.; Jones, C. T.	2018	Leveling the Joint Task Force Core Competencies for Clinical Research Professionals.	Therapeutic Innovation and Regulatory Science	**Joint Task Force for Clinical Trial Competency (JTF) Core Competecies Framework 2.0**. Use 3 levels of competencies. They are **(1) Fundamental** ("Can perform the task/and or exhibit the knowledge at an essential or fundamental level; may require some coaching or supervision"), **(2) Skilled** ("Can perform task or skill independently, consistently, accurately, and has a moderate level of expertise. Efficient and high-quality work; able to independently navigate resources and uses tools well"), and **(3) Advanced** ("Demonstrates advanced skills and knowledge and the ability to teach, coach, or supervise others. Consistently applies critical thinking and problem solving".	The JTF Framework is composed of **8 Domains (Figure 1) 1- Scientific Concepts and Research Design 2- Ethical and Participant Safety Considerations 3- Investigational Products Development and Regulation 4- Study Operations (Good Clinical Practice) 5- Study and Site Management 6 Data Management and Informatics 7-Leadership and Professionalism 8-Communications and Teamwork- and 47 core competency statements** that were aligned and harmonized utilizing published statements of core competency requirements to encompass the entire clinical research enterprise.	Individual	Ultimately, the profession of clinical research was recognized by the Commission on Accreditation of Allied Health Education Programs (CAAHEP). academic programs in clinical research are now eligible for CAAHEP accreditation based on JTF core competency-informed educational standards.	Applicable to all research roles	n/a	2016	CB	n/a
Swenson-Britt, E.; Berndt, A.	2013	Development and Psychometric Testing of the Nursing Research Self-Efficacy Scale (NURSES).	Journal of Nursing Measurement	**Nursing Research Self-Efficacy Scale (NURSES)** is a 38-item instrument using a **5-point Likert scale** format ranging from 1 (very little) to 5 (quite a lot). Time to complete the NURSES instrument would be about 15 min.	Measure individual nurses' degree of research self-efficacy and their perceptions regarding their unit's collective support of research use. The NURSES is a 38-**item Likert scale instrument** developed through ongoing instrument validation that included content validation and exploratory and confirmatory analysis. The instrument contained **5 subscales** include obtaining science-based knowledge resources, critically reading and evaluating quantitative research literature, critically reading and evaluating qualitative research literature, understanding and applying theory, and collective research efficacy.	Individual	An innovative conceptual framework, synthesizing organizational culture and leadership. This model is presented as a theoretical subscription using a modified Gibb's Framework (McQuiston & Campbell, 1997), theory (Schein, 2004) with social cognitive theory (Bandura, 1997), was the basis for this instrument development.	Nurses from multiple hospitals.	The total sample included 687 nurses who responded to the survey; of these nurses, 649 completed all sections out of the 2000 contacted.	In 4 phases between 2005 and 2008	Research Self-Efficacy	Swenson-Britt & Berndt, (2009) used the term research self-efficacy to describe the extent to which practicing nurses expressed confidence in their ability to understand and use research findings. Specifically, RSE measured the extent to which practicing nurses could (a) obtain research resources; (b) understand theory; (c) evaluate quantitative research; (d) evaluate qualitative research; and (e) apply research findings in their practice.
Torres, G. C. S.; Estrada, M. G; Sumile, E. F. R.; Macindo, J. R. B.;Maravilla, S. N.; Hendrix, C. C.	2017	Assessment of Research Capacity Among Nursing Faculty in a Clinical Intensive University in The Philippines.	Nursing Forum	**Modified University of Memphis (UM) questionnaire** contained **30 items** to measure the ability and motivation to do research among faculty members and reportedly took about 15 min to complete. **Of the roginal 41 items** some items were revised, and non relevant deleted. Lastly, the study team added new items to include self-assessment of knowledge and skill on the research process, including research designs, and items to assess perceived resources and barriers in conducting research at UST-CON. Perceived levels of research capacity were assessed using a **Likert scale, with 1 as the lowest possible score and 5 as the highest.**	**5 AREAS- 30 subareas:** subject demographics; 1- Conceptualization phase (5 subcategorie) 2- Design and planning phase (10 subcategories) 3- Empirical phase (6 Sub) - 4- Analytic phase (5 sub) 5 - Dissemination phase (4 sub)	Individual	Meyers, A., & Kurtz, H. (2013). University of Memphis research capacity assessment taskforce report. Retrieved October 21, 2014, from http://www.memphis.edu/researchsupport/pdfs/research capacity_assessment.pdf	Clinical and nonclinical faculty members who had a regular teaching assignment at UST-CON.	Of the 82 faculty at UST-CON, 66 participated (80.49%) in the study. Thirty faculty (45.45%) completed the survey electronically, while the rest chose the pen-and-paper method. faculty members.	The survey was conducted from August to October 2015.	Research capacity	Research capacity is the capability to conduct high-quality research undertakings and is crucial toward building the knowledge base for evidence based nursing practice (Sergott, McIvor, & Green, 2006).
Williams, C.; Miyazaki, K.; Borkowski, D.; McKinstry, C.; Cotchet, M.; Haines, T.	2015	Research capacity and culture of the Victorian public health allied health workforce is influenced by key research support staff and location	Australian Health Review	**The research capacity and culture (RCC) tool. 15 questions for Individal level (Table 4).** Each item used a 10-point numeric rating scale response format with 1 being the lowest and 10 being the highest success or skill leve	The tool contains **51 items examining participants' self-perceived skills or success in a range of areas related to research capacity or culture at the individual, team or organisational level.** Each item used a 10-point numeric rating scale response format with 1 being the lowest and 10 being the highest success or skill level.Aforced response format was used throughout the questionnaire to reduce missing data and the 'don't know' option was removed.	Individual, team and organisation levels	SELECTED PAPER ABOVE: Holden L, Pager S, Golenko X, Ware RS. Validation of the research capacity and culture (RCC) tool: measuring RCC at individual, team and organisation levels. Aust J Primary Health 2012; 18: 62–7. doi:10.1071/ PY10081	Allied health professions invited to participate were dietetics, drug and alcohol workers, exercise physiology, music therapy, occupational therapy, oral health (not dentistry), orthotics and prosthetics, physiotherapy, play therapy, podiatry,psychology, social work, speech pathology, audiology, orthoptics, pathology, pharmacy, radiation therapy, radiography and sonography.	Of the 539 allied health participantswhostarted the online survey, 520 fully completed the surveys.	The survey was open from 10 April to 1 June 2014.	n/a	n/a

Figure 2 - *PRISMA screening diagram*



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3/4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4/16
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6/7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7/8
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	8
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	10
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	11
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	9
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	n/a
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	14
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/a
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10/20/21
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	14
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	14
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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